Higher-Order Asymptotics of Regression-Adjusted Estimators in Randomized Trials

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Abstract

Regression adjustment is widely used to improve the precision of average treatment effect estimates from randomized controlled trials. Interacted regression weakly dominates simple regression in terms of its first-order behavior but, in 50-50 randomized trials, the asymptotic variance of both methods is the same and so relative performance is determined by higher-order behavior. Here, we conduct a higher order analysis to compare simple vs. interacted regression adjustments in the equal-probability case. The relative performance of the methods depends on a number of factors including feature skewness, treatment-effect heterogeneity, heteroskedasticity, and nonlinear misspecification—and can favor either estimator. However, in some practically important settings, including when features are skewed and outlying feature values are associated with higher outcome noise, simple regression can outperform interacted regression by a material amount.

1 Introduction

Randomized controlled trials (RCTs) are

fundamental to how causal effects are ascertained across a number of fields. One valuable attribute of RCTs is that they enable unbiased estimation of average treatment effects using simple methods such as the difference in means, without reliance on parametric assumptions [Imbens and Rubin, 2015]. However, while the difference-in-means estimator is unbiased, this does not automatically make it the preferred approach; and in particular regression adjustments are often used to improve power. For example, the Food and Drug Administration issued recent guidance encouraging analysts to "adjust for baseline covariates that are strongly associated with an outcome in a clinical trial population" [U.S. Food and Drug Administration, 2023, accompanying guidance snapshot].

The study of regression adjustments in RCTs has a long tradition in statistics. Fisher [1935] used arguments rooted in parametric linear modeling to advocate for such regression adjustments. Fisher's analysis was later critiqued by Freedman [2008], who argued that reliance on linear models was misaligned with model-free guarantees sought in RCT inference. Lin [2013], however, responded to Freedman's critique by showing that regression adjustments do not require parametric assumptions to be near-unbiased for the average treatment effect.

Code, simulation scripts and the computational certificates used in the proofs: https://github.com/kelleyan-pixel/higher-order-regression-adjustment-for-rct.

Claude Opus 5.5 was used for code generation, including numerical simulation and computational proof design via SymPy [Meurer et al., 2017]; GPT-6 Astra was used for checking proofs and writing assistance throughout the appendix.

Furthermore, Lin [2013] showed that regression adjustments with full treatment-control interaction weakly dominate both the difference in means estimator and Fisher's simple regression adjustment in terms of their leading-order (root-n scale) errors. Lin's weak dominance result has led such treatment-interacted regressions to be often presented as a safe starting point for use of regression adjustments in RCTs [Li and Ding, 2017, Negi and Wooldridge, 2021, Wager, 2025].

A gap in this narrative, however, is that treatment-interacted regressions only strictly dominate simple regression in designs with randomization probability $\pi \neq 0.5$. In evenly randomized trials with $\pi = 0.5$, the first-order accuracy of both methods is equal for any data-generating distribution, i.e., first-order theory offers no guidance on which method to prefer. And, while $\pi = 0.5$ is of course a special case, this is an important special case in applications, and it is not a priori clear we should prefer interacted regression to simple regression here. For example, Zhao and Ding [2023] argue based on simulation evidence and heuristics on the number of free parameters that simple regression may in practice be more stable than interacted regression with limited sample size—especially when there are multiple treatment levels. Just to give a concrete example, suppose that

$$\log(X_i) \sim \mathcal{N}\left(0,\, 1.1^2\right), \quad Y_i(w) = 0.9X_i + w + (1 + X_i)U_i, \quad U_i \sim \mathcal{N}\left(0,\, 1\right). \tag{1}$$

Then, at sample size $n = 500$, interacted regression has 7% higher mean-squared error for the average treatment effect than simple regression.

Motivated by this gap, we here conduct a higher-order asymptotic analysis of both simple and interacted regression adjustments in RCTs to elucidate relative merits of both approaches in the case where $\pi = 0.5$. We frame our findings in terms of asymptotic deficiency [Hodges and Lehmann, 1970]. When two estimators have matching first-order behavior but one dominates the other on the second-order scale then, asymptotically, switching to the better one gives an accuracy boost that is equivalent to a fixed boost in sample size, which is referred to as the deficiency. Our main finding is that, while either the simple or interacted regression adjustment may have better second-order performance, this trade-off appears to favor simple regression in some realistic settings.

As a practical extension of this theory, we study feasible inference choices for both estimators with heteroskedasticity-robust standard error estimates. In particular, we consider both interacted and simple regression point estimates in tandem with HC1 and jackknife/HC3 standard error estimates [Efron and Stein, 1981, MacKinnon and White, 1985]. For both point estimates, HC1 can undercover substantially under skewed, tail-heteroskedastic covariates. The jackknife improves coverage substantially in most designs but is still liable to undercover in cases of rare outliers or large HTE. In a typical low-HTE regime, simple regression combined with the jackknife consistently achieves at least nominal coverage with the smallest interval length.

1.1 Related Work

Regression adjustment in randomized experiments. The literature on regression adjustment in randomized experiments sits in two camps. The design-based literature treats the potential outcomes and covariates as fixed in a finite population, with randomness coming solely from treatment assignment. In this setting, Lin [2013] shows that interacted regression weakly dominates both simple regression and difference-in-means in asymptotic efficiency. Meanwhile, Zhao and Ding [2023] compare their finite sample behavior and allow for multi-armed and factorial experiments, postulating that interacted regression exhibits poor finite sample performance due to its additional model complexity. Another interesting but more distant line of

work, including Lei and Ding [2021] and Chiang et al. [2025], addresses bias and inference of the interacted estimator as p grows with n.

By contrast, the superpopulation framework treats the experimental units as random draws from a larger population and studies repeated sampling over both the units and treatment assignment. Most past work here has focused on first-order comparisons. For instance, Negi and Wooldridge [2021] show that the interacted regression estimator weakly dominates both difference in means and simple regression in this framework as well. We continue this line of work to formally investigate higher-order effects of both estimators, in particular to resolve the ambiguous optimality at $\pi = 0.5$. Importantly, our goal is to characterize the higher-order behavior of the two estimators under IID sampling specifically, and it is possible that a design-based analysis yields different second-order results.

Higher-order asymptotic theory. First-order theory inherently cannot compare estimators with the same asymptotic variance, even though their finite-sample behavior may differ meaningfully. To address this, Nagar [1959] introduced expansions of the higher-order moments of estimators, and Sargan [1974] gave conditions for their validity; meanwhile Edgeworth expansions approximate their underlying distribution [Sargan, 1976, Bhattacharya and Rao, 1976, Hall, 1992, Ghosh, 1994]. As we focus on the mean-squared error, we primarily build on the former tools, although the latter are useful for rigorous results on tail behavior and inference. Hodges and Lehmann [1970] introduced asymptotic deficiency as a measure of the additional sample size required by the less accurate of two first-order equivalent procedures, and Rothenberg [1984] emphasized that higher-order approximations should be consulted when competing estimators have the same limiting distribution. Parallel to our contribution, Newey and Smith [2004] present a higher-order comparison of the first-order equivalent general method of moments and generalized empirical likelihood estimators. Tan [2014] studies the second-order properties of regression and likelihood calibration estimators in survey sampling and missing-data problems, finding that neither consistently dominates. Chiang et al. [2025] derive second-order stochastic expansions of regression-adjusted estimators to compare inference in a design-based, growing p regime. However, to our knowledge we are the first to compare interacted and simple regression in a superpopulation setting using higher-order expansions.

Variance estimation. Under superpopulation sampling, the fixed-design based conventional sandwich variance for the interacted estimator ignores the variance of the sample covariate mean used for centering [Imbens and Wooldridge, 2009, Section 5.3]; see also Lin [2013, Section 5]. Imbens and Wooldridge [2009] add an estimate of the omitted term, and Negi and Wooldridge [2021] derive the asymptotic variance that accounts for it; we take this result for granted. General finite-sample behavior of heteroskedasticity-robust variance estimators is studied by Efron and Stein [1981], MacKinnon and White [1985], Chesher and Jewitt [1987], Long and Ervin [2000] and Imbens and Kolesár [2016]; the HC* nomenclature was introduced in MacKinnon and White [1985]. It is well known that despite their popularity in applied research, HC0 and HC1 are prone to undercovering in finite samples, and the statistics literature typically recommends switching to the jackknife (or HC3 in the MacKinnon and White [1985] terminology). Lei and Ding [2021] and Chiang et al. [2025] demonstrate this point specifically in the case of regression adjustment inference. Following this work, we present an empirical study of feasible inference using regression adjustment combined with robust standard errors, but we do not claim to contribute to this literature from a theoretical standpoint.

2 Asymptotics of Regression Adjustment

We focus on completely randomized trials (CRTs) with causal effects formalized using the potential outcomes model [Imbens and Rubin, 2015]. We assume the covariates $X_i \in \mathbb{R}^p$ and potential outcomes $Y_i(0), Y_i(1) \in \mathbb{R}$ are drawn IID from a distribution P, with average treatment effect $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$. Both regression estimators below include an intercept, and interacted regression is evaluated at the sample mean of the covariates, so translating the covariates leaves both estimators unchanged; we therefore assume without loss of generality that $\mathbb{E}[X] = 0$. Treatment is assigned randomly conditional on the fixed treated group size $n_1 = \sum_{i=1}^n W_i = \lfloor \pi n \rfloor$; treatment assignments are exchangeable but not independent. The analyst observes (X_i, Y_i, W_i) for $i = 1, \ldots, n$, where $Y_i = Y_i(W_i)$. We consider the following three estimators of the ATE:

$$\begin{aligned}
\hat{\tau}_{DM} &= \frac{1}{|\{i : W_i = 1\}|} \sum_{\{i:W_i=1\}} Y_i - \frac{1}{|\{i : W_i = 0\}|} \sum_{\{i:W_i=0\}} Y_i, \\
\hat{\tau}_{REG} &= \hat{\tau}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau, \\
\hat{\tau}_{IREG} &= \hat{\tau} + \frac{1}{n}\sum_{i=1}^n X_i\hat{\delta}, \text{ with } Y_i \sim \alpha + X_i\beta + W_i\tau + W_iX_i\delta,
\end{aligned} \tag{2}$$

where, e.g., $Y_i \sim \alpha + X_i\beta + W_i\tau$ means we algorithmically run a regression of Y_i against X_i and W_i with an intercept (without making any claims on well-specification of such a regression). To describe their first-order behavior it is helpful to define the best linear approximation to the conditional response function of the potential outcomes,

$$\left\{\alpha^*_{(w)},\ \beta^*_{(w)}\right\} = \underset{\alpha_{(w)},\,\beta_{(w)}}{\operatorname{argmin}} \left\{\mathbb{E}\left[\left(Y_i(w) - X_i\beta_{(w)} - \alpha_{(w)}\right)^2\right]\right\}, \tag{3}$$

together with the oracle residuals

$$\varepsilon_i(w) = Y_i(w) - \alpha^*_{(w)} - X_i^\top\beta^*_{(w)}, \tag{4}$$

Writing $Z_i = \Sigma^{-1/2}X_i$ with $\Sigma = \operatorname{Var}[X_i]$, we have $\mathbb{E}[Z_i] = 0$, $\mathbb{E}[Z_iZ_i^\top] = I_p$, $\mathbb{E}[Z_i\varepsilon_i(w)] = 0$ and $\varepsilon_i = \varepsilon_i(W_i)$. The first-order behavior of the three estimators is well known [Negi and Wooldridge, 2021].

Proposition 1. *Suppose that X_i and $Y_i(w)$ have finite second moments and that $\operatorname{Var}[X_i]$ is invertible. Then all three estimators above are asymptotically normal, $\sqrt{n}(\hat{\tau}_\cdot - \tau) \Rightarrow \mathcal{N}(0, V_\cdot)$, with limiting variances*

$$\begin{aligned}
V_{DM} &= V_{IREG} + \operatorname{Var}\left[X_i\left(\sqrt{\frac{\pi}{1-\pi}}\,\beta^*_{(0)} + \sqrt{\frac{1-\pi}{\pi}}\,\beta^*_{(1)}\right)\right], \\
V_{REG} &= V_{IREG} + \frac{(1-2\pi)^2}{\pi(1-\pi)}\operatorname{Var}\left[X_i\left(\beta^*_{(1)} - \beta^*_{(0)}\right)\right], \\
V_{IREG} &= \operatorname{Var}\left[X_i\left(\beta^*_{(1)} - \beta^*_{(0)}\right)\right] + \sum_{w=0,\,1}\frac{1}{\mathbb{P}[W_i = w]}\mathbb{E}\left[\left(Y_i(w) - X_i\beta^*_{(w)} - \alpha^*_{(w)}\right)^2\right].
\end{aligned} \tag{5}$$

Thus $\hat{\tau}_{IREG}$ weakly dominates both $\hat{\tau}_{DM}$ and $\hat{\tau}_{REG}$ at every π; its advantage over $\hat{\tau}_{REG}$ is strict for generic distributions P when $\pi \neq 0.5$, while at $\pi = 0.5$ the two regression estimators have the same asymptotic variance

$$V^* = \operatorname{Var}\left[X_i\left(\beta^*_{(1)} - \beta^*_{(0)}\right)\right] + 2\mathbb{E}\left[\varepsilon_i^2(1)\right] + 2\mathbb{E}\left[\varepsilon_i^2(0)\right]. \tag{6}$$

This is the setting we study. Because first-order theory is indifferent between the two methods, we turn to second-order asymptotics.

Importantly, the true MSE may not converge as $n \to \infty$ if the estimators have divergent higher moments. To address this issue, we consider a capped mean-squared error. For a fixed $c > 0$, define

$$\overline{\mathrm{MSE}}\left(\hat{\tau}\right) = \mathbb{E}\left[\left(\hat{\tau} - \tau\right)^2 \wedge c^2\right] \tag{7}$$

This updated loss agrees with the regular squared error whenever $|\hat{\tau} - \tau| \leq c$ while bounding the contribution of unstable regression fits. We use the same fixed cutoff for both estimators. For this risk calculation, set an estimator to zero whenever its least-squares fit is not uniquely defined, including an empty treatment arm. Any measurable finite-valued completion gives the same expansion below. The cap is a loss convention depending on the target τ, not a modification of the estimator.

The proof of the following result is given in Appendix A. Appendix B gives simulation-based comparisons of realized, uncapped MSE with the formula in a variety of settings.

Theorem 2. *Consider a CRT on n units with $n_1 = \lfloor n/2 \rfloor$, and suppose that $(X_i, Y_i(0), Y_i(1))$ are IID from a distribution P, $\Sigma = \mathrm{Var}\left[X_i\right] \succ 0$, and*

$$\mathbb{E}\big[\|X_i\|^{12}\big],\ \mathbb{E}\big[|Y_i(w)|^{12}\big] < \infty \text{ for } w = 0,\, 1.$$

Define the whitened features and corresponding effect-heterogeneity coefficients as

$$Z_i = \Sigma^{-1/2} X_i, \quad \Sigma = \mathrm{Var}\left[X_i\right], \qquad \gamma = \Sigma^{1/2}\left(\beta^*_{(1)} - \beta^*_{(0)}\right), \tag{8}$$

and let

$$K = \mathbb{E}[ZZ^\top \|Z\|^2], \qquad S = \mathbb{E}[Z\|Z\|^2], \qquad S_\gamma = \mathbb{E}[ZZ^\top (Z^\top \gamma)],$$
$$\mathcal{E} = \mathbb{E}[ZZ^\top \varepsilon], \qquad \mathcal{E}_W = \mathbb{E}[ZZ^\top (W - \tfrac{1}{2})\varepsilon] = \tfrac{1}{4}\big(\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]\big).$$

Here expectations involving W use an auxiliary Bernoulli$(1/2)$ variable independent of the units, with $\varepsilon = \varepsilon(W)$. Then

$$\begin{aligned}
&\overline{\mathrm{MSE}}\left(\hat{\tau}_{IREG}\right) - \overline{\mathrm{MSE}}\left(\hat{\tau}_{REG}\right) = \frac{D_p}{n^2} + O(n^{-3}), \\
&D_p = -(2p+3)\|\gamma\|^2 + \gamma^\top K \gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2 \\
&\qquad - 16\,\mathrm{tr}(S_\gamma \mathcal{E}_W) - 8(\gamma^\top S)\,\mathrm{tr}(\mathcal{E}_W) + 8\gamma^\top \mathcal{E}_W S + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2].
\end{aligned} \tag{9}$$

Consequently, when $V^* > 0$, the asymptotic deficiency of IREG relative to REG, computed from the capped MSE, is $\Delta = D_p / V^*$ with V^* as in (6); the sample-size interpolation used here is specified in Appendix A.

Remark 3 (Bias and variance contributions). One may naturally be interested in how the overall MSE difference decomposes into bias and variance. Writing $e_{j,c} = \max\{-c, \min\{\hat{\tau}_j - \tau, c\}\}$, define

$$\mathrm{Bias}_j = \mathbb{E}[e_{j,c}], \qquad \mathrm{Var}_j = \mathrm{Var}\left[e_{j,c}\right], \qquad \overline{\mathrm{MSE}}(\hat{\tau}_j) = \mathrm{Var}_j + \mathrm{Bias}_j^2.$$

Evaluating the leading bias term for the two estimators (Lemma 14 in Appendix A) gives

$$\mathrm{Bias}_{\mathrm{REG}} = -\frac{4\,\mathrm{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}), \qquad \mathrm{Bias}_{\mathrm{IREG}} = -\frac{4\,\mathrm{tr}(\mathcal{E}_W)}{n} + O(n^{-2}),$$

so that $n^2\{\mathrm{Bias}^2_{\mathrm{IREG}} - \mathrm{Bias}^2_{\mathrm{REG}}\} = -(\gamma^\top S)^2 - 8(\gamma^\top S)\,\mathrm{tr}(\mathcal{E}_W)$ to leading order, indeed directly reproducing two terms; the remaining terms are due to variance.

The expression for D_p combines four key blocks: a covariate shape block scaled by the linear HTE γ; terms crossing γ with residual asymmetry; residual heteroskedasticity; and nonlinear misspecification. No single term determines the sign and either method can come up on top—even when the population linear slopes coincide (i.e., when $\gamma = 0$). When $\gamma = 0$, the expression in (9) reduces to

$$D_p = 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2],$$

and while $\mathrm{tr}(\mathcal{E}^2) = \|\mathcal{E}\|_F^2 \geq 0$, the term $S^\top \mathbb{E}[Z\varepsilon^2]$ can have either sign, so Δ can be positive or negative even when $\gamma = 0$. To interpret the scale of Δ write

$$R_\tau^2 = \frac{\|\gamma\|^2}{\|\gamma\|^2 + 4\mathbb{E}[\varepsilon_i^2]} = \frac{\|\gamma\|^2}{V^*},$$

which, when $V^* > 0$, quantifies the share of the first-order variance V^* attributable to linear treatment-effect heterogeneity.

Corollary 4. *Under the conditions of Theorem 2, suppose furthermore that the potential outcomes satisfy a well-specified linear model with homoskedastic errors,* $\mathbb{E}\left[\varepsilon_i(w) \mid X_i\right] = 0$ *and* $\mathbb{E}\left[\varepsilon_i^2(w) \mid X_i\right] = \sigma_w^2$ *for* $w \in \{0,1\}$. *Then* $\mathcal{E} = \mathcal{E}_W = 0$ *and* $\mathbb{E}[Z\varepsilon^2] = 0$. *For* $\gamma \neq 0$,

$$\Delta = R_\tau^2 \left[-(2p+3) + \frac{\gamma^\top K\gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2}\right]. \tag{10}$$

In the Gaussian sub-case $K = (p+2)I_p$, $S = 0$ *and* $S_\gamma = 0$, *so that* $\Delta = -(p+1)R_\tau^2$. *If* $\gamma = 0$, *the second-order MSE difference vanishes for any covariate distribution; when* $V^* > 0$, $R_\tau^2 = 0$ *and* $\Delta = 0$.

In the Gaussian homoskedastic case, the second-order coefficient favors IREG and grows linearly in the number of covariates, with p fixed as $n \to \infty$. This complicates the parameter-count heuristic of Zhao and Ding [2023], showing that the finite sample advantage of either estimator cannot be reduced to any simple feature of the DGP.

However, the following corollary sheds some light on why REG has been observed to outperform IREG to perform worse in many real-life datasets, or in examples such as (1).

Corollary 5. *Under the conditions of Theorem 2, suppose furthermore that the potential outcomes follow*

$$Y_i(w) = \alpha_{(w)}^* + X_i^\top \beta_{(w)}^* + \eta_i \|Z_i\|,$$

where $\mathbb{E}\left[\eta_i\right] = 0$, $\mathbb{E}\left[\eta_i^2\right] = \nu^2$ *and* η_i *is independent of* (Z_i, W_i). *Then* $\mathbb{E}\left[\varepsilon_i(w) \mid Z_i\right] = 0$ *and* $\mathbb{E}\left[\varepsilon_i^2(w) \mid Z_i\right] = \nu^2\|Z_i\|^2$, *where* $\|\cdot\|$ *is the Euclidean norm, so that* $\mathbb{E}\left[Z_i\varepsilon_i^2(w)\right] = \nu^2 S$ *and* $\mathbb{E}\left[\varepsilon_i^2\right] = \nu^2 p$. *For* $\gamma \neq 0$,

$$\Delta = R_\tau^2 \left[-(2p+3) + \frac{\gamma^\top K\gamma - (\gamma^\top S)^2 - 3\|S_\gamma\|_F^2}{\|\gamma\|^2}\right] + \frac{8\nu^2\, S^\top S}{\|\gamma\|^2 + 4\nu^2 p}. \tag{11}$$

When $\gamma = 0$ *and* $\nu^2 > 0$, *the deficiency is* $\Delta = 2\|S\|^2/p$, *which is nonnegative and is strictly positive whenever* $S \neq 0$.

It is useful to consider how the terms in (11) compare in practice. In many applications the linear HTE signal is small relative to the residual departure from the regression model, so R_τ^2 is small and the first summand rarely dominates unless its bracket is large. The bracketed component is of order p in the leading cases—it equals $-(p+1)$ under Gaussian covariates—and

its moment terms grow with the kurtosis and skewness of the underlying covariates, so it is sensitive to departures from Gaussianity in either direction.

The remaining term is nonnegative, since $8\nu^2\|S\|^2/(\|\gamma\|^2 + 4\nu^2 p) \geq 0$, and for $\nu > 0$ it vanishes if and only if $S = \mathbb{E}[Z\|Z\|^2] = 0$. Right-skewed covariates with a long upper tail, such as measures of user activity, engagement or spending, can produce a large third-moment vector S after centering and whitening, in which case this term can dominate the first summand. The scalar case makes this transparent: with $p = 1$, Corollary 5 gives

$$\Delta = R_\tau^2 \left\{ \mathbb{E}\left[Z_i^4\right] - 4\mathbb{E}\left[Z_i^3\right]^2 - 5 \right\} + \frac{8\nu^2 \mathbb{E}\left[Z_i^3\right]^2}{\gamma^2 + 4\nu^2}.$$

Using Pearson's inequality $\mathbb{E}\left[Z_i^4\right] \geq \mathbb{E}\left[Z_i^3\right]^2 + 1$ [see, e.g., Rohatgi and Székely, 1989] and $8\nu^2/(\gamma^2 + 4\nu^2) = 2(1 - R_\tau^2)$,

$$\Delta \geq -4R_\tau^2 + (2 - 5R_\tau^2)\mathbb{E}\left[Z_i^3\right]^2.$$

So when $R_\tau^2 < 2/5$ the deficiency is positive as soon as $\mathbb{E}\left[Z_i^3\right]^2 > 4R_\tau^2/(2 - 5R_\tau^2)$, and the lower bound grows linearly in $\mathbb{E}\left[Z_i^3\right]^2$.

3 Feasible Inference

While the above results concern the precision of the point estimates provided by REG and IREG, in practice we are also interested in the validity of the resulting confidence intervals—which now depends jointly on how we estimate the standard error. As before, we operate in an assumption-lean framework, only considering estimators that are robust to misspecification of the underlying linear model.

We fit REG and IREG as in the R package `estimatr` [Blair et al., 2024]: REG is `lm_robust(y ~ w + x)`, and IREG is `lm_lin(y ~ w, covariates = ~ x)`, with `se_type = "HC1"` (the package default is HC2). Because we operate in a superpopulation framework while these implementations are for fixed design sampling, we add a plug-in estimate of the variance of the sample covariate mean used for centering (Imbens and Wooldridge, 2009, Section 5.3; Negi and Wooldridge, 2021):

$$\hat{G} = \hat{\delta}^\top \hat{\Sigma}_X \hat{\delta}, \qquad \widehat{\mathrm{Var}}_{\mathrm{SP}}(\hat{\tau}_{\mathrm{IREG}}) = \widehat{\mathrm{Var}}_{\mathrm{HC1}}(\hat{\tau}_{\mathrm{IREG}}) + \frac{\hat{G}}{n}, \tag{12}$$

where $\hat{\delta}$ is the vector of interaction coefficients and $\hat{\Sigma}_X$ the sample covariance of the covariates; $\hat{G}$ estimates $G = \|\gamma\|^2$. Because $\hat{\delta}$ is itself noisy, $\hat{G}$ is biased upward, by about $\mathrm{tr}\{\mathrm{Var}(\hat{\delta})\,\Sigma\}$.

We choose to investigate HC1 as a popular benchmark: it is Stata's default and used in 98.1% of 40,571 robust Stata regressions in reproduction packages of leading economics journals [Kranz, 2024]. We compare it with a leverage-aware alternative, the delete-one jackknife applied to the final estimate [MacKinnon and White, 1985]. Let $\hat{\tau}_{(-i)}$ be the estimator recomputed on the sample with unit i removed—both arm-specific fits are refitted and, for IREG, the covariates are recentred at the leave-one-out mean $\bar{X}_{(-i)}$—the jackknife variance estimate is then given by

$$\hat{V}_{\mathrm{JK}} = \frac{n-1}{n} \sum_{i=1}^{n} \left(\hat{\tau}_{(-i)} - \bar{\hat{\tau}}\right)^2, \qquad \bar{\hat{\tau}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\tau}_{(-i)}. \tag{13}$$

Because the jackknife operates on the final treatment effect estimate, it accounts for the sampling variation of the covariate mean and thus requires no superpopulation correction. Thus

		IREG					REG				
			HC1 + plug-in		jackknife			HC1		jackknife	
Design	R^2_τ	RMSE	width	cov.	width	cov.	RMSE	width	cov.	width	cov.
Small or no heterogeneity											
Gaussian	0.022	0.091	0.36	95.0	0.36	95.0	0.091	0.36	94.8	0.36	95.0
Misspecified, $G = 0$	0	0.098	0.38	94.9	0.38	95.0	0.098	0.38	94.8	0.38	95.0
Skewed, het., $G = 0$	0	0.167	0.65	95.0	0.66	95.4	0.165	0.64	95.0	0.65	95.5
Heavy-tailed, het.	0.016	0.175	0.64	94.7	0.66	95.5	0.168	0.63	94.7	0.64	95.5
Lognormal $s = 1.2$, $g = 0.3$	0.013	0.425	1.52	94.6	1.57	95.6	0.408	1.47	94.6	1.52	95.7
Lognormal $s = 1.5$, $g = 0.1$	0.002	0.798	2.55	94.7	2.73	96.2	0.715	2.40	94.9	2.55	96.7
Lognormal $s = 1.5$, $g = 0.5$	0.049	0.839	2.62	93.7	2.79	95.3	0.753	2.46	93.2	2.61	95.1
Lognormal $s = 1.8$, $g = 0.3$, $\sigma = 1 + 1.5X$	0.009	2.826	7.26	94.3	8.04	96.4	2.196	6.57	94.3	7.18	96.6
Lognormal $s = 2$, $g = 0.1$	0.002	3.433	7.76	94.8	8.78	97.0	2.458	6.80	95.0	7.55	97.6
Moderate heterogeneity											
Lognormal $s = 1.5$, $g = 1$	0.172	0.901	2.81	92.0	2.97	93.7	0.826	2.64	90.2	2.81	91.9
Rare extreme values	0.2	0.100	0.40	94.0	0.40	94.0	0.102	0.40	94.7	0.41	95.5
Large heterogeneity											
Heavy-tailed, het.	0.365	0.220	0.75	94.2	0.77	94.6	0.212	0.73	91.9	0.76	93.0
Lognormal $s = 1.5$, $g = 3$	0.651	1.451	4.46	89.8	4.57	90.4	1.380	4.17	84.1	4.51	86.0
Gaussian	0.692	0.161	0.64	95.1	0.64	95.1	0.162	0.64	95.0	0.64	95.2

Table 1: Root-mean-squared error of the point estimators, and median width and coverage (%) of nominal 95% intervals for the population ATE: IREG with HC1 plus the plug-in correction and with the direct jackknife (13), and REG with HC1 and with the direct jackknife ($n = 500$, 20,000 replications; "het." denotes heteroskedastic noise). Designs are defined in Appendix D; g is the treatment-effect slope of the scalar lognormal designs. RMSE is computed over replications where both point estimators exist; coverage counts unavailable intervals as failures, and median width is computed over available intervals. In the rare-value design, 6.3% of replications have a leverage-one unit in an IREG arm, whose leave-one-out estimate is omitted from the jackknife (Section 3).

for IREG we report HC1 with the plug-in correction (12) alongside the jackknife; for REG, we directly report HC1 and jackknife. When removing unit i leaves a rank-deficient design (unit i has leverage one in its fit), $\hat{\tau}_{(-i)}$ is undefined; we then omit it and apply (13) to the remaining n' leave-one-out estimates, with n' in place of n. All intervals use t critical values with $n - k$ degrees of freedom, where k is the number of regression coefficients.

Table 1 compares all four joint procedure possibilities across 14 DGPs. These DGPs exhibit a range of values of R^2_τ and covariate distributions, including Gaussian, skewed, heavy-tailed, and rare-value covariates. Appendix D defines each design. All have $n = 500$ and 20,000 replications, and every procedure is applied to the same simulated datasets.

When R^2_τ is small (< 0.05), the combination of REG and jackknife is a clear winner across our selected designs. Its 95% coverage ranges from 95.0% to 97.6%, its median interval width is between 0.86 and 1.00 times that of IREG with the jackknife and between 0.97 and 1.00 times that of IREG with HC1 plus the plug-in, which covers 93.7–95.0%. REG is also at least as precise a point estimator in these designs, with MSE ratios IREG/REG between 1.00 and 1.95. At the 95% level both estimators achieve only slightly anticonservative coverage with HC1; for IREG the upward bias of the plug-in partly offsets HC1's understatement of variance, and the undercoverage shows up more at other nominal levels (Table 2). The price of the jackknife is width: under the heaviest tails it is conservative, and REG+jackknife intervals are up to 1.11 times as wide as REG+HC1's.

This pattern does not hold across all DGPs. As R^2_τ grows with skewed covariates, REG undercovers with either variance estimator, down to 86.0% with jackknife at $R^2_\tau = 0.651$, while

Design	R^2_τ	IREG, HC1 + plug-in 80	95	99	IREG, jackknife 80	95	99	REG, HC1 80	95	99	REG, jackknife 80	95	99
Gaussian	0.022	80.1	95.0	99.1	80.2	95.0	99.1	79.9	94.8	99.0	80.2	95.0	99.0
Misspecified, $G = 0$	0	79.8	94.9	99.0	79.8	95.0	99.0	79.6	94.8	98.9	79.9	95.0	99.0
Skewed, het., $G = 0$	0	80.1	95.0	99.0	80.8	95.4	99.1	80.0	95.0	98.9	80.8	95.5	99.1
Heavy-tailed, het.	0.016	79.1	94.7	98.9	80.8	95.5	99.1	79.0	94.7	98.9	80.9	95.5	99.1
Lognormal $s = 1.2$, $g = 0.3$	0.013	78.4	94.6	98.9	80.6	95.6	99.2	78.6	94.6	98.9	80.8	95.7	99.2
Lognormal $s = 1.5$, $g = 0.1$	0.002	77.0	94.7	99.3	81.2	96.2	99.6	77.8	94.9	99.3	82.5	96.7	99.6
Lognormal $s = 1.5$, $g = 0.5$	0.049	76.6	93.7	98.5	80.4	95.3	98.9	76.4	93.2	98.3	80.6	95.1	98.8
Lognormal $s = 1.8$, $g = 0.3$, $\sigma = 1 + 1.5X$	0.009	75.5	94.3	99.1	81.5	96.4	99.5	76.4	94.3	99.0	83.1	96.6	99.5
Lognormal $s = 2$, $g = 0.1$	0.002	75.1	94.8	99.3	82.3	97.0	99.6	76.1	95.0	99.3	84.2	97.6	99.7

Table 2: Coverage (%) at nominal 80%, 95% and 99% in the designs of Table 1 with small or no heterogeneity, for IREG with HC1 plus the plug-in and with the jackknife, and for REG with HC1 and with the jackknife.

IREG with jackknife is the most reliable procedure, although it also reaches only 90.4% there. In the rare-value design an IREG arm contains a single rare unit in 6.3% of replications; removing that unit leaves a rank-deficient fit, so its LOO estimate is omitted, and IREG+jackknife covers 89.9% of those refits against 94.3% of the others. Treating any undefined leave-one-out estimate as making the entire interval unavailable would instead give IREG+jackknife coverage of 88.4%. REG is essentially unaffected: removing a unit leaves its pooled fit rank deficient in only 0.03% of replications.

Table 2 tests this pattern in the low-R^2_τ DGPs across nominal levels. HC1 intervals are close to nominal at 95% and 99% but undercover at 80% in the skewed designs, by up to 4.9 percentage points for IREG with the plug-in and 3.9 for REG, so their 95% coverage is partly a crossing point of a miscalibrated interval. With the jackknife, neither estimator is more than 0.2 points below nominal at any of the three levels in any of these designs; under the heaviest tails it is conservative, with REG+jackknife covering up to 84.2% at nominal 80%.

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A Proofs

The goal of this section is to derive the constant D_p in Theorem 2. The argument has three parts. First, we express a completely randomized trial as two independent samples and obtain an MSE expansion for a fixed treatment fraction. Second, we calculate the second-order constant under Bernoulli assignment. Third, we remove the effect of random treatment imbalance to recover the constant for a balanced trial. The remaining lemmas translate the answer into matrix notation and establish the identities used in the numerical checks.

Throughout this appendix, $j \in \{\mathrm{REG}, \mathrm{IREG}\}$ identifies the estimator and $c > 0$ is the fixed loss cutoff in (7). The subscripts Bern and (CRT, r) indicate, respectively, independent Bernoulli(1/2) assignment and complete randomization with $m = rn$ treated units. Expectations under these designs are written $\mathbb{E}_{\mathrm{Bern}}$ and $\mathbb{E}_{\mathrm{CRT},r}$. In particular,

$$\overline{\mathrm{MSE}}_{\mathcal{D}}(\hat{\tau}_j) = \mathbb{E}_{\mathcal{D}}[\min\{(\hat{\tau}_j - \tau)^2, c^2\}], \qquad \mathcal{D} \in \{\mathrm{Bern}, (\mathrm{CRT}, r)\}.$$

An omitted fraction in a CRT subscript means $r = \lfloor n/2 \rfloor / n$. Estimators use the finite-valued completion specified after (7).

Definition 1 (Taylor-polynomial MSE)**.** Let $\bar{\mathcal{V}} = n^{-1} \sum_{i=1}^{n} \mathcal{V}_i$ and $\hat{\tau} = g(\bar{\mathcal{V}})$ locally, where g is smooth near $\mu = \mathbb{E}[\mathcal{V}_i]$ and $g(\mu) = \tau$. Put $d = \bar{\mathcal{V}} - \mu$, $a_i = \partial_i g(\mu)$, $H_{ij} = \partial_{ij} g(\mu)$ and $A_{3,ijk} = \partial_{ijk} g(\mu)$. Repeated coordinate indices in this definition are summed from 1 to the dimension of $\mathcal{V}_i$. The cubic Taylor polynomial is

$$\tilde{\tau}_n = \tau + a_i d_i + \tfrac{1}{2} H_{ij} d_i d_j + \tfrac{1}{6} A_{3,ijk} d_i d_j d_k, \tag{14}$$

and its mean-squared error is defined by

$$\mathrm{MSE}^{\exp}(\hat{\tau}) := \mathbb{E}[(\tilde{\tau}_n - \tau)^2].$$

The superscript exp labels the Taylor expansion; this is the exact squared-error expectation of the polynomial, not of $\hat{\tau}$ itself. It depends on the coefficient map and raw-moment coordinates specified below.

For fixed $r = m/n \in (0,1)$, let $U_i^{(w)}$ contain X_i, $Y_i(w)$, the distinct entries of $X_i X_i^\top$, and $X_i Y_i(w)$, and put $\mu_w = \mathbb{E}[U_i^{(w)}]$. Let $\bar{U}_w$ be its sample mean in arm w. The local coefficient map is $\hat{\tau}_j = h_{j,r}(\bar{U}_1, \bar{U}_0)$, given explicitly in Lemma 6. Write $\mathcal{P}_{j,r}$ for its cubic Taylor polynomial at $\mu = (\mu_1, \mu_0)$, using the displayed derivative formula with $d = (\bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$, and define

$$\mathrm{MSE}^{\exp}_{\mathrm{CRT},r}(\hat{\tau}_j) = \mathbb{E}_{\mathrm{CRT},r}[(\mathcal{P}_{j,r} - \tau)^2].$$

We abbreviate $\mathcal{P}_{j,r}$ to $\mathcal{P}_r$ when the estimator j is fixed.

For Bernoulli assignment use the IID vector $\mathcal{V}_i = (W_i, W_i U_i^{(1)}, (1-W_i)U_i^{(0)})$, with mean $\mu = (1/2, \mu_1/2, \mu_0/2)$. For its three blocks $v = (v_W, v_1, v_0)$, the local coefficient map is

$$g_j(v) = h_{j,v_W}\big(v_1/v_W, v_0/(1-v_W)\big).$$

Let $\tilde{\tau}_{n,j}$ be the cubic polynomial (14) for this map and define

$$\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_j) = \mathbb{E}_{\mathrm{Bern}}[(\tilde{\tau}_{n,j} - \tau)^2].$$

The maps need only be defined near their population means; their Taylor polynomials are defined for every sample-moment vector, including samples on which a regression fit is undefined.

Lemma 7 shows that Taylor-polynomial MSE agrees with the capped MSE through order n^{-2}, with an $O(n^{-3})$ difference, under the sampling and moment conditions of Theorem 2. No inverse-Gram moment assumption is needed for this capped loss. Ordinary, uncapped MSE requires additional tail control. For example, if the completed estimator has finite second moment and

$$\|\hat{\tau}_j - \tilde{\tau}_{n,j}\|_{L^2} = o(n^{-3/2}),$$

under Bernoulli assignment, then $\|\tilde{\tau}_{n,j} - \tau\|_{L^2} = O(n^{-1/2})$ and Cauchy–Schwarz give an $o(n^{-2})$ difference between its ordinary MSE and $\mathrm{MSE}^{\exp}$. This sufficient condition is not implied by the population moment assumptions. For fixed-fraction CRT, the same sufficient condition uses $\mathcal{P}_{j,r}$ in place of $\tilde{\tau}_{n,j}$.

Statements about deficiency require $V^* > 0$. We interpret a noninteger sample-size shift by linear interpolation of the capped risks between even sample sizes. On that sequence, each risk has the form $V^*/n + c_j(1/2)/n^2 + O(n^{-3})$, where $c_j(r)$ is the second-order coefficient defined by Lemma 8, and hence

$$\overline{\mathrm{MSE}}_{\mathrm{IREG}}(n+\Delta) - \overline{\mathrm{MSE}}_{\mathrm{REG}}(n) = O(n^{-3}), \qquad \Delta = \frac{c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2)}{V^*}.$$

Here $\overline{\mathrm{MSE}}_j(n)$ denotes the capped risk of estimator j at balanced sample size n, and noninteger arguments denote its linear interpolation. An integer implementation entails rounding. For odd n, the individual n^{-2} risk coefficients can have a common parity correction; the same-n difference in Theorem 2 is unchanged through that order. If $V^* = 0$, the risk-difference identity still holds, but division by V^* does not define a deficiency.

Population expectations involving W in the second-order constants are evaluated at the nominal allocation: an auxiliary Bernoulli(1/2) variable independent of the potential outcomes

and covariates. This convention also applies when the realized CRT has odd n, with treatment fraction $\lfloor n/2 \rfloor / n$. The effect of that $O(n^{-1})$ imbalance on the same-n risk difference is included in the remainder.

We use $G = \|\gamma\|^2$. In covariate-moment tensor formulas, indices such as i, j, k, l, a, b, c, d range from 1 to p and refer to covariate coordinates, rather than experimental units. Repeated indices are summed over unless stated otherwise. In particular, δ_{jk} is the Kronecker delta. Indices of the raw-moment vector $\mathcal{V}$ instead range over the coordinates of that vector; an estimator subscript $j \in \{\mathrm{REG}, \mathrm{IREG}\}$ is never summed.

A.1 Two independent samples and a fixed-fraction expansion

The first lemma explains why superpopulation sampling simplifies the complete-randomization calculation. Once the treatment assignment is fixed, the two arms consist of disjoint sets of independent units.

We use the arm raw-moment vectors and their means from Definition 1.

Lemma 6 (Two-sample representation)**.** *Assume the sampling and moment conditions of Theorem 2. Let the assignment vector W be independent of the sampled units, with $\sum_i W_i = m$ and $0 < m < n$. Conditionally on W, the treated observations $\{(X_i, Y_i(1)) : W_i = 1\}$ and the control observations $\{(X_i, Y_i(0)) : W_i = 0\}$ are independent IID samples of sizes m and $n - m$ from the laws of $(X, Y(1))$ and $(X, Y(0))$ respectively.*

Furthermore, wherever the required fits are uniquely defined, both $\hat{\tau}_{\mathrm{REG}}$ and $\hat{\tau}_{\mathrm{IREG}}$ are of the form $h_{j,r}(\bar{U}_1, \bar{U}_0)$ with $r = m/n$, where $h_{j,r}$ is infinitely differentiable in a neighborhood of (μ_1, μ_0) and its derivatives at (μ_1, μ_0) are rational functions of r without poles in $(0, 1)$.

Proof. Conditioning on W fixes two disjoint index sets. Independence of W and the sampled units leaves the unit distributions unchanged; independence across units then gives independence both within and between the arms.

To establish the second claim, note that the coordinates of $\bar{U}_w$ determine all per-arm means, second moments, and cross-moments needed for least squares. Both estimators are rational functions of these coordinates. To see this explicitly, let $C_w = \overline{XX^\top}_w - \bar{X}_w \bar{X}_w^\top$ and $s_w = \overline{XY}_w - \bar{X}_w \bar{Y}_w$. Define the arm-specific and pooled within-arm slopes by

$$\hat{\beta}_w = C_w^{-1} s_w, \qquad \hat{\beta}_{\mathrm{pool}}(r) = \{rC_1 + (1-r)C_0\}^{-1}\{rs_1 + (1-r)s_0\}.$$

Writing $d_X = \bar{X}_1 - \bar{X}_0$, the two coefficient maps are

$$h_{\mathrm{REG},r}(\bar{U}_1, \bar{U}_0) = \bar{Y}_1 - \bar{Y}_0 - d_X^\top \hat{\beta}_{\mathrm{pool}}(r),$$
$$h_{\mathrm{IREG},r}(\bar{U}_1, \bar{U}_0) = \bar{Y}_1 - \bar{Y}_0 - d_X^\top \{(1-r)\hat{\beta}_1 + r\hat{\beta}_0\}.$$

The second formula evaluates each arm's fitted regression at $\bar{X} = r\bar{X}_1 + (1-r)\bar{X}_0$. At (μ_1, μ_0), $d_X = 0$, both maps equal τ, and each population covariance matrix is $\Sigma \succ 0$. This gives smoothness and the stated dependence on r. □

Lemma 7 (Localization for capped loss and clipped errors)**.** *Under the sampling and moment conditions of Theorem 2, for any fixed $c > 0$ and $\eta \in (0, 1/2)$,*

$$\overline{\mathrm{MSE}}_{\mathrm{CRT},r}(\hat{\tau}_j) = \mathrm{MSE}^{\mathrm{exp}}_{\mathrm{CRT},r}(\hat{\tau}_j) + O(n^{-3}), \tag{15}$$

uniformly for $r = m/n \in [1/2 - \eta, 1/2 + \eta]$. Also, with $e_{j,c} = \max\{-c, \min\{\hat{\tau}_j - \tau, c\}\}$,

$$\mathbb{E}[e_{j,c}] = \mathbb{E}[\mathcal{P}_r - \tau] + O(n^{-2}) \tag{16}$$

uniformly on the same interval. The analogous statements hold under independent Bernoulli$(1/2)$ *assignment with the unit-level cubic Taylor polynomial* $\tilde{\tau}_{n,j}$.

Proof. Each coordinate of $U^{(w)}$ has a finite sixth moment: for example,

$$\mathbb{E}|X_k Y(w)|^6 \le \{\mathbb{E}|X_k|^{12}\,\mathbb{E}|Y(w)|^{12}\}^{1/2},$$

and the same argument applies to $X_k X_l$. For $\mu = (\mu_1, \mu_0)$ and $d = (\bar{U}_1 - \mu_1, \bar{U}_0 - \mu_0)$, centered sample-mean moment counting and independence of the arms give, uniformly over these fractions,

$$\mathbb{E}\|d\|^6 = O(n^{-3}), \qquad \mathbb{E}\|d\|^4 = O(n^{-2}).$$

Indeed, in any product of centered observations a sample index appearing only once has zero expectation. At most three distinct indices can therefore contribute to a degree-six moment. The same count shows that the expectation of every signed degree-five monomial in d is $O(n^{-3})$; an absolute fifth-moment bound alone would not give this order.

By Lemma 6 and compactness of the fraction interval, there is a fixed $\delta > 0$ such that on $\mathcal{A} = \{\|d\| \le \delta\}$ both regression maps are smooth, their derivatives through order six are uniformly bounded, and $|h_{j,r}(\mu + d) - \tau| \le c$. Thus the cap is inactive there and $\Pr(\mathcal{A}^c) = O(n^{-3})$. Write q_k for the homogeneous degree-k Taylor term of $h_{j,r} - \tau$. Taylor expansion of the squared error through degree five gives, on $\mathcal{A}$,

$$(h_{j,r} - \tau)^2 = q_1^2 + 2q_1q_2 + q_2^2 + 2q_1q_3 + 2q_1q_4 + 2q_2q_3 + O(\|d\|^6).$$

For every $0 \le k \le 6$,

$$\mathbb{E}[\|d\|^k \mathbf{1}_{\mathcal{A}^c}] \le \delta^{k-6}\mathbb{E}\|d\|^6 = O(n^{-3}).$$

Consequently each displayed polynomial can be integrated over the whole sample space with only $O(n^{-3})$ error. The two degree-five terms have signed expectations $O(n^{-3})$, and $\mathbb{E}[q_3^2] = O(n^{-3})$. Comparing with $\mathbb{E}[(q_1 + q_2 + q_3)^2]$, and bounding the capped loss on $\mathcal{A}^c$ by c^2, proves (15). This argument also shows why any finite-valued completion on undefined fits gives the same expansion: all such fits lie outside $\mathcal{A}$.

For clipped bias, the clipping is again inactive on $\mathcal{A}$. The cubic Taylor remainder there is $O(\|d\|^4)$ and hence has expectation $O(n^{-2})$. The clipped error is bounded by c off $\mathcal{A}$, and the off-event expectations of the cubic polynomial are $O(n^{-3})$ by the preceding bound. This proves (16).

Under Bernoulli assignment one may use $\mathcal{V}_i = (W_i, W_i U_i^{(1)}, (1 - W_i)U_i^{(0)})$. These IID raw-moment vectors have finite sixth moments. At their population mean both treatment probabilities are $1/2$ and the population Gram matrices are invertible. The same argument applies to the smooth coefficient maps in these coordinates, with $d = \bar{\mathcal{V}} - \mathbb{E}\mathcal{V}$. □

Lemma 8 (Conditional expansion)**.** *Under the sampling and moment conditions of Theorem 2, let* $\mathcal{P}_r$ *be the third-order Taylor polynomial of* $h_{j,r}$ *at* (μ_1, μ_0) *and define* $\mathrm{MSE}^{\exp}_{\mathrm{CRT},r}(\hat{\tau}_j) = \mathbb{E}[(\mathcal{P}_r - \tau)^2]$. *For every* $\eta \in (0, \frac{1}{2})$,

$$\mathrm{MSE}^{\exp}_{\mathrm{CRT},r}(\hat{\tau}_j) = \frac{V_j(r)}{n} + \frac{c_j(r)}{n^2} + O(n^{-3}), \qquad j \in \{\mathrm{REG}, \mathrm{IREG}\}, \tag{17}$$

uniformly in $r = m/n \in [\frac{1}{2} - \eta, \frac{1}{2} + \eta]$, *with* V_j *and* c_j *infinitely differentiable in* r. *The same expansion holds with* $\overline{\mathrm{MSE}}_{\mathrm{CRT},r}$ *in place of* $\mathrm{MSE}^{\exp}_{\mathrm{CRT},r}$.

Proof. Fix an estimator $j \in \{\mathrm{REG}, \mathrm{IREG}\}$ and an interval $I_\eta = [\frac{1}{2} - \eta, \frac{1}{2} + \eta]$ with $0 < \eta < 1/2$. The arm sizes are $m = rn$ and $n - m = (1-r)n$. We first expand the squared Taylor polynomial, then determine which sample-mean moments can contribute at orders n^{-1} and n^{-2}.

Step 1: expand the squared estimation error. Write $d_1 = \bar{U}_1 - \mu_1$ and $d_0 = \bar{U}_0 - \mu_0$. Then $h_{j,r}(\mu_1, \mu_0) = \tau$, and the third-order Taylor polynomial of $h_{j,r}$ at (μ_1, μ_0) can be written

$$\mathcal{P}_r - \tau = q_1 + q_2 + q_3,$$

where q_k is a polynomial of degree k in the coordinates of (d_1, d_0) whose coefficients are the kth derivatives of $h_{j,r}$ at (μ_1, μ_0) divided by $k!$. By Lemma 6, each of these coefficients is a rational function of r without poles in $(0, 1)$. Expanding the square,

$$\mathrm{MSE}^{\mathrm{exp}}_{\mathrm{CRT},r}(\hat{\tau}_j) = \mathbb{E}[q_1^2] + 2\mathbb{E}[q_1 q_2] + \big(\mathbb{E}[q_2^2] + 2\mathbb{E}[q_1 q_3]\big) + 2\mathbb{E}[q_2 q_3] + \mathbb{E}[q_3^2].$$

Step 2: justify the required moments and separate the arms. After expanding the polynomials, each expectation is a linear combination of mixed moments of the form

$$\mathbb{E}\Big[\prod_{u=1}^{a} d_{1,k_u} \prod_{u=1}^{b} d_{0,l_u}\Big], \qquad 2 \le a + b \le 6, \tag{18}$$

with coefficients that are products of two Taylor coefficients, and hence rational in r without poles in $(0, 1)$. Each coordinate of $U^{(w)}$ is one of X_k, $Y(w)$, $X_k X_l$ or $X_k Y(w)$. By the Cauchy–Schwarz inequality, $\mathbb{E}|X_k X_l|^6 \le (\mathbb{E}|X_k|^{12}\,\mathbb{E}|X_l|^{12})^{1/2}$, and similarly for $X_k Y(w)$, so the sampling and moment conditions of Theorem 2 give $\mathbb{E}\|U^{(w)}\|^6 < \infty$. By Hölder's inequality, every mixed moment of order at most six of the coordinates of $U^{(w)}$ is then finite.

By Lemma 6, conditionally on W the treated and control observations are independent IID samples of sizes m and $n - m$. In particular d_1 and d_0 are independent, and (18) factors as

$$\mathbb{E}\Big[\prod_{u=1}^{a} d_{1,k_u}\Big]\,\mathbb{E}\Big[\prod_{u=1}^{b} d_{0,l_u}\Big].$$

Step 3: count powers of the arm size. Consider the treated-arm factor. The control-arm calculation is the same, with $U^{(0)}$ and $n - m$ in place of $U^{(1)}$ and m. Write $\kappa_{k_1 \cdots k_s}$ for the joint cumulant of $(U^{(1)}_{k_1}, \ldots, U^{(1)}_{k_s})$. By the IID property,

$$\mathrm{cum}(d_{1,k_1}, \ldots, d_{1,k_s}) = m^{-s} \sum_{i : W_i = 1} \mathrm{cum}\big(U^{(1)}_{i,k_1}, \ldots, U^{(1)}_{i,k_s}\big) = m^{1-s}\,\kappa_{k_1 \cdots k_s} \qquad (2 \le s \le 6),$$

while the first-order cumulants of d_1 are zero. The moment–cumulant formula writes $\mathbb{E}[\prod_{u=1}^{a} d_{1,k_u}]$ as a sum over set partitions π of $\{1, \ldots, a\}$ of $\prod_{B \in \pi} \mathrm{cum}(d_{1,k_u} : u \in B)$. A partition with a singleton block contributes zero. A partition into blocks of size at least two contributes a product of cumulants of $U^{(1)}$ times $m^{\sum_{B \in \pi}(1 - |B|)} = m^{-(a - |\pi|)}$, and such a partition has $1 \le |\pi| \le \lfloor a/2 \rfloor$. Hence, for $2 \le a \le 6$,

$$\mathbb{E}\Big[\prod_{u=1}^{a} d_{1,k_u}\Big] = \sum_{s = \lceil a/2 \rceil}^{a-1} \frac{M_s^{(1)}}{m^s}, \tag{19}$$

where each $M_s^{(1)}$ is a finite sum of products of cumulants of $U^{(1)}$ and does not depend on m; the moment equals 0 for $a = 1$ and 1 for $a = 0$. For example,

$$\mathbb{E}[d_{1,k} d_{1,l}] = \frac{\kappa_{kl}}{m}, \qquad \mathbb{E}[d_{1,k} d_{1,l} d_{1,q}] = \frac{\kappa_{klq}}{m^2},$$
$$\mathbb{E}[d_{1,k} d_{1,l} d_{1,q} d_{1,v}] = \frac{\kappa_{kl}\kappa_{qv} + \kappa_{kq}\kappa_{lv} + \kappa_{kv}\kappa_{lq}}{m^2} + \frac{\kappa_{klqv}}{m^3}.$$

These identities hold exactly for every m; no approximation is involved.

Step 4: collecting powers of n. Substitute $m = rn$ and $n - m = (1-r)n$. By (19), a mixed moment (18) with a treated and b control coordinates vanishes if $a = 1$ or $b = 1$, and otherwise is a finite sum of terms

$$\frac{M_s^{(1)} M_t^{(0)}}{r^s (1-r)^t}\, n^{-(s+t)}, \qquad \lceil a/2 \rceil \le s, \quad \lceil b/2 \rceil \le t,$$

with $s = 0$ and $M_0^{(1)} = 1$ when $a = 0$, and $t = 0$ and $M_0^{(0)} = 1$ when $b = 0$. Its power of $1/n$ is therefore at least $\lceil a/2 \rceil + \lceil b/2 \rceil \ge \lceil (a+b)/2 \rceil$, and at most 5. Combining these moment identities with the squared expansion in Step 1,

$$\mathrm{MSE}^{\exp}_{\mathrm{CRT},r}(\hat{\tau}_j) = \sum_{\ell=1}^{5} \frac{e_\ell(r)}{n^\ell},$$

where each e_ℓ is a finite sum of products of Taylor coefficients, cumulants of $U^{(1)}$ and $U^{(0)}$, and powers of r^{-1} and $(1-r)^{-1}$. Each e_ℓ is thus rational in r without poles in $(0,1)$, and in particular infinitely differentiable and bounded on I_η. The expansion is exact: a polynomial in $1/n$, not an asymptotic series.

The coefficient of n^{-2} does not require every term in the square. The following degree count identifies the terms that matter. The degree-two term $\mathbb{E}[q_1^2]$ contributes to $\ell = 1$ only:

$$\mathbb{E}[q_1^2] = \frac{g_1^\top \operatorname{Var}\left[U^{(1)}\right] g_1}{m} + \frac{g_0^\top \operatorname{Var}\left[U^{(0)}\right] g_0}{n-m} = \frac{1}{n}\left\{ \frac{g_1^\top \operatorname{Var}\left[U^{(1)}\right] g_1}{r} + \frac{g_0^\top \operatorname{Var}\left[U^{(0)}\right] g_0}{1-r} \right\},$$

where g_w is the gradient of $h_{j,r}$ with respect to $\bar{U}_w$ at (μ_1, μ_0). The degree-three term $2\mathbb{E}[q_1 q_2]$ contributes to $\ell = 2$ only, through third cumulants. The degree-four terms $\mathbb{E}[q_2^2] + 2\mathbb{E}[q_1 q_3]$ contribute to $\ell = 2$ and $\ell = 3$. The degree-five and degree-six terms contribute only to $\ell \ge 3$. Hence

$$V_j(r) = e_1(r), \qquad c_j(r) = e_2(r),$$

where $c_j(r)$ collects the n^{-2} parts of $2\mathbb{E}[q_1 q_2]$, $\mathbb{E}[q_2^2]$ and $2\mathbb{E}[q_1 q_3]$.

Step 5: make the remainder bound uniform. Because r stays away from 0 and 1, the remaining coefficients are bounded independently of the treatment fraction. Specifically, for every $n \ge 1$ and admissible $r \in I_\eta$,

$$\left| \mathrm{MSE}^{\exp}_{\mathrm{CRT},r}(\hat{\tau}_j) - \frac{V_j(r)}{n} - \frac{c_j(r)}{n^2} \right| = \left| \sum_{\ell=3}^{5} \frac{e_\ell(r)}{n^\ell} \right| \le \frac{1}{n^3} \sum_{\ell=3}^{5} \sup_{r' \in I_\eta} |e_\ell(r')|.$$

The right-hand side is finite because each e_ℓ is continuous on the compact interval I_η. This is the claimed $O(n^{-3})$ bound, uniformly in $r \in I_\eta$. Lemma 7 transfers it to the capped risk. □

A.2 From Bernoulli assignment to balanced complete randomization

Bernoulli assignment makes the unit-level moment calculation convenient, but it also makes the treatment fraction random. The next lemma permits us to average the fixed-fraction expansion over that random fraction. The theorem that follows identifies the correction for fixing the arm sizes instead.

Lemma 9 (Bernoulli and conditional expansions)**.** *Under the sampling and moment conditions of Theorem 2, with* $W_i \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ *independent of the units, let* $\bar{W} = n^{-1}\sum_i W_i$*. For every* $\eta \in (0, 1/2)$*,*

$$\overline{\text{MSE}}_{\text{Bern}}(\hat{\tau}_j) = \mathbb{E}\Big[\overline{\text{MSE}}_{\text{CRT},\bar{W}}(\hat{\tau}_j)\mathbf{1}\{|\bar{W} - \tfrac{1}{2}| \le \eta\}\Big] + O(e^{-2n\eta^2}).$$

Consequently,

$$\text{MSE}^{\text{exp}}_{\text{Bern}}(\hat{\tau}_j) = \mathbb{E}\Big[\text{MSE}^{\text{exp}}_{\text{CRT},\bar{W}}(\hat{\tau}_j)\mathbf{1}\{|\bar{W} - \tfrac{1}{2}| \le \eta\}\Big] + O(n^{-3}).$$

The indicated conditional risks and polynomials are used only on the central event, where both arms are nonempty for large n; each integrand containing its indicator is defined as zero off that event. The fixed-fraction and Bernoulli Taylor polynomials need not coincide conditionally: the second identity follows by transferring both to the capped risks, not by identifying the polynomials.

Proof. Conditional on $n\bar{W} = m$, Bernoulli assignment is uniform over assignments with m treated units, exactly as in a CRT. The law of total expectation therefore identifies the conditional capped risk exactly. The omitted event contributes at most

$$c^2 \Pr(|\bar{W} - 1/2| > \eta) \le 2c^2 e^{-2n\eta^2}.$$

Applying Lemma 7 to the Bernoulli risk and uniformly to the conditional risks proves the polynomial identity with $O(n^{-3})$ error. □

The proof of Theorem 2 combines these lemmas, a reduction from the CRT to the Bernoulli design, and the Bernoulli calculation.

Theorem 10. *Suppose the Bernoulli design has second-order difference*

$$\text{MSE}^{\text{exp}}_{\text{Bern}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}^{\text{exp}}_{\text{Bern}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}}}{n^2} + O(n^{-3}) \tag{20}$$

at $\pi = 1/2$*, and let the sampling and moment conditions of Theorem 2 hold. Then, under complete randomization with* $n_1 = \lfloor n/2 \rfloor$ *treated units,*

$$\text{MSE}^{\text{exp}}_{\text{CRT}}(\hat{\tau}_{\text{IREG}}) - \text{MSE}^{\text{exp}}_{\text{CRT}}(\hat{\tau}_{\text{REG}}) = \frac{D_{\text{Bern}} + 4G}{n^2} + O(n^{-3}). \tag{21}$$

The same identity holds for the capped MSE.

Proof. The correction comes from treatment imbalance: Bernoulli assignment occasionally produces unequal arms, where IREG has a first-order advantage. Averaging this advantage contributes $-4G/n^2$ to the Bernoulli comparison. A balanced trial removes that contribution.

Step 1: average the first-order variance difference. Let $n_1 = \sum_{i=1}^n W_i$ under the Bernoulli design and write $r = n_1/n$. Conditional on $n_1 = m$, the treatment assignment has the same distribution as a CRT with treatment fraction r. Fix $\eta \in (0, \frac{1}{2})$ and write $\mathcal{A}_\eta = \{|r - 1/2| \le \eta\}$. All expectations of functions containing r^{-1} or $(1-r)^{-1}$ below are restricted to $\mathcal{A}_\eta$; their products with the indicator are defined as zero off that event. By Lemma 9, $\text{MSE}^{\text{exp}}_{\text{Bern}}$ is, up to $O(n^{-3})$, the average over n_1 of $\text{MSE}^{\text{exp}}_{\text{CRT},r}$ restricted to $|r - \frac{1}{2}| \le \eta$, and by Lemma 8 the latter admits the expansion (17) uniformly on that set, with V_j and c_j infinitely differentiable. At treatment fraction r the first-order variance difference is, by Proposition 1,

$$V_{\text{IREG}}(r) - V_{\text{REG}}(r) = -\frac{(1-2r)^2}{r(1-r)}G,$$

and

$$\frac{(1-2r)^2}{r(1-r)} = 16u^2 + O(u^4), \qquad u = r - \tfrac{1}{2}.$$

Under Bernoulli randomization with $\pi = 1/2$, $\mathbb{E}[u^2] = 1/(4n)$ and $\mathbb{E}[u^4] = O(n^{-2})$. Restricting these bounded moments to $\mathcal{A}_\eta$ changes them by an exponentially small amount. The event is symmetric under $u \mapsto -u$, so $\mathbb{E}[u\mathbf{1}_{\mathcal{A}_\eta}] = 0$. It follows that

$$\frac{1}{n}\mathbb{E}\big[\{V_{\mathrm{IREG}}(r) - V_{\mathrm{REG}}(r)\}\mathbf{1}_{\mathcal{A}_\eta}\big] = -\frac{4G}{n^2} + O(n^{-3}).$$

Step 2: average the second-order coefficient. For the second-order coefficients, a second-order Taylor expansion of $c_{\mathrm{IREG}} - c_{\mathrm{REG}}$ around $1/2$ together with the restricted moment identities just established gives

$$\mathbb{E}[\{c_{\mathrm{IREG}}(r) - c_{\mathrm{REG}}(r)\}\mathbf{1}_{\mathcal{A}_\eta}] = c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) + O(n^{-1}).$$

Averaging (17) over n_1 therefore yields

$$\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{REG}}) = \frac{c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) - 4G}{n^2} + O(n^{-3}),$$

so

$$D_{\mathrm{Bern}} = c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) - 4G.$$

Step 3: fix the arm sizes. Under balanced complete randomization, $r = 1/2$ for even n, giving

$$D_{\mathrm{CRT}} = c_{\mathrm{IREG}}(1/2) - c_{\mathrm{REG}}(1/2) = D_{\mathrm{Bern}} + 4G.$$

For even n, Lemma 8 therefore gives directly

$$\mathrm{MSE}^{\exp}_{\mathrm{CRT}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{CRT}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_{\mathrm{Bern}} + 4G}{n^2} + O(n^{-3}).$$

For odd n, $r - 1/2 = O(n^{-1})$. The first-order variance difference is quadratic in $r - 1/2$, so its contribution to the MSE is $O(n^{-3})$; the smoothness of $c_{\mathrm{IREG}} - c_{\mathrm{REG}}$ gives an $O(n^{-1})$ change in the second-order coefficient, which is also $O(n^{-3})$ after division by n^2. Thus the same expansion holds for odd n. Lemma 7 gives the capped-risk conclusion. □

A.3 The second-order constant under Bernoulli assignment

We now calculate D_{Bern}. The tensor notation below is used only to organize the contractions in the Taylor expansion; Lemma 12 converts the result to the matrix notation of the main theorem. Here F captures a fourth-moment contraction, Q and $\mathcal{R}$ capture third-moment contractions, and A_P, B_P, C_P combine third moments with arm-specific residual structure.

Theorem 11. *Let the sampling and moment conditions of Theorem 2 hold, except that* $W_i \overset{\mathrm{iid}}{\sim}$ *Bernoulli*$(1/2)$ *independently of* $(X_i, Y_i(0), Y_i(1))$. *Define*

$$T_{3,ijk} = \mathbb{E}[Z_i Z_j Z_k], \quad T_{4,ijkl} = \mathbb{E}[Z_i Z_j Z_k Z_l], \quad q_i = T_{3,ijk}\delta_{jk}, \quad L_{ij} = T_{3,iab}T_{3,jab}, \tag{22}$$

write $S = q = \mathbb{E}[Z\|Z\|^2]$, *and set* $\mathcal{E} = \mathbb{E}[ZZ^\top\varepsilon]$ *and* $P = \mathcal{E}_W = \mathbb{E}[ZZ^\top(W - \frac{1}{2})\varepsilon]$. *Here* P *denotes this residual-moment matrix, not the superpopulation law. For the scalar contractions*

$$G = \|\gamma\|^2, \qquad F = \gamma_i\gamma_j T_{4,ijkk}, \qquad Q = (\gamma^\top q)^2, \qquad \mathcal{R} = \gamma^\top L\gamma, \tag{23}$$

$$A_P = \gamma_i T_{3,ijk} P_{jk}, \qquad B_P = (\gamma^\top q)\,\mathrm{tr}(P), \qquad C_P = \gamma^\top P q, \tag{24}$$

we have

$$\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_{\mathrm{REG}}) = \frac{D_{\mathrm{Bern}}}{n^2} + O(n^{-3}), \tag{25}$$

where

$$D_{\mathrm{Bern}} = -(2p+7)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]. \tag{26}$$

The same expansion holds for the capped MSE.

Proof. Step 1: remove population main effects and whiten the covariates. On the neighborhood used in Lemma 7, both REG and IREG contain the regressors 1, W and X and their fits are uniquely defined. Subtracting any linear combination of these regressors from the outcome changes the fitted coefficient on W by the corresponding amount while leaving the estimation error unchanged, so we may subtract the population main effects and the term τW_i and work with the exact representation

$$Y_i = W_i X_i^\top \{\beta^*_{(1)} - \beta^*_{(0)}\} + \varepsilon_i.$$

This local invariance determines the Taylor coefficients; any failure of equivariance in the completion affects capped risk by only $O(n^{-3})$. Whitening as in (8) gives the canonical representation $Y_i = W_i \gamma^\top Z_i + \varepsilon_i$, whose population ATE is zero. In the following calculation, Y_i denotes this transformed outcome, $g(\mu) = 0$, and the resulting Taylor errors equal those for the original estimators about their original target τ. The raw-moment transformation is affine, so the Taylor-polynomial risks in Definition 1 are preserved. We write

$$W_i = \tfrac{1}{2} + \omega_i, \qquad \omega_i \in \{-\tfrac{1}{2}, \tfrac{1}{2}\}, \qquad \mathbb{E}[\omega_i] = 0, \qquad \omega_i^2 = \tfrac{1}{4}.$$

Step 2: identify the three second-order contributions. Use the Bernoulli raw-moment vector $\mathcal{V}_i$ in Definition 1, formed with the transformed outcome and whitened covariates from Step 1, and the corresponding map g_j. Suppress the estimator subscript temporarily, so that $\hat{\tau} = g(\bar{\mathcal{V}})$ locally and $\tilde{\tau}_n$ is given by (14). Let

$$\Sigma^{\mathcal{V}}_{ij} = \mathrm{Cov}(\mathcal{V}_{1,i}, \mathcal{V}_{1,j}), \qquad \mu^{\mathcal{V}}_{3,ijk} = \mathbb{E}[(\mathcal{V}_{1,i} - \mu_i)(\mathcal{V}_{1,j} - \mu_j)(\mathcal{V}_{1,k} - \mu_k)]$$

be the covariance and third central-moment tensors of one unit's raw-moment vector; the first subscript in $\mathcal{V}_{1,i}$ denotes the unit and the second denotes its coordinate. In the next three contractions all repeated indices range over raw-moment coordinates. Expanding the square of (14) gives

$$n\,\mathrm{MSE}^{\exp}_{\mathrm{Bern}}(\hat{\tau}_j) = V^* + \frac{\mathcal{T}_{1,j} + \mathcal{T}_{2,j} + \mathcal{T}_{3,j}}{n} + O(n^{-2}),$$

where, suppressing j again,

$$\mathcal{T}_1 = a_i H_{jk} \mu^{\mathcal{V}}_{3,ijk}, \tag{27}$$

$$\mathcal{T}_2 = \frac{1}{4} H_{ij} H_{kl} \left(\Sigma^{\mathcal{V}}_{ik}\Sigma^{\mathcal{V}}_{jl} + \Sigma^{\mathcal{V}}_{il}\Sigma^{\mathcal{V}}_{jk} + \Sigma^{\mathcal{V}}_{ij}\Sigma^{\mathcal{V}}_{kl}\right), \tag{28}$$

$$\mathcal{T}_3 = \frac{1}{3} a_i A_{3,jkl} \left(\Sigma^{\mathcal{V}}_{ij}\Sigma^{\mathcal{V}}_{kl} + \Sigma^{\mathcal{V}}_{ik}\Sigma^{\mathcal{V}}_{jl} + \Sigma^{\mathcal{V}}_{il}\Sigma^{\mathcal{V}}_{jk}\right), \tag{29}$$

The omitted terms are $O(n^{-2})$ in $n\,\mathrm{MSE}^{\exp}$: the non-pairing part of the fourth moment and all degree-five and degree-six products in the squared polynomial contribute $O(n^{-3})$ to $\mathrm{MSE}^{\exp}$, by the same centered sample-mean moment count as in Lemma 8. Lemma 7 gives the same $O(n^{-3})$ remainder for the capped MSE. The difference between the order-n^{-2} MSE coefficients of REG and IREG is obtained by evaluating these three contractions for the two estimators.

The three contributions have a simple origin: $\mathcal{T}_1$ is the product of the linear and quadratic Taylor terms, $\mathcal{T}_2$ is the square of the quadratic term, and $\mathcal{T}_3$ is the product of the linear and cubic terms. Thus the calculation requires derivatives only through order three.

Step 3: compute derivatives from the least-squares normal equations. For REG, order the regressors as $(1, Z, W)$; for IREG, use $(1, Z, W, WZ)$. In these orders, the population Gram matrices and their inverses in the whitened coordinates are

$$A_R = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & I_p & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}, \qquad A_R^{-1} = \begin{pmatrix} 2 & 0 & -2 \\ 0 & I_p & 0 \\ -2 & 0 & 4 \end{pmatrix}, \qquad A_I = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & I_p & 0 & I_p/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & I_p/2 & 0 & I_p/2 \end{pmatrix},$$

$$A_I^{-1} = \begin{pmatrix} 2 & 0 & -2 & 0 \\ 0 & 2I_p & 0 & -2I_p \\ -2 & 0 & 4 & 0 \\ 0 & -2I_p & 0 & 4I_p \end{pmatrix}, \qquad \theta_R = (0, \gamma/2, 0), \qquad \theta_I = (0, 0, 0, \gamma).$$

For either regressor vector, let $A(v)$ and $b(v)$ be its Gram matrix and regressor–outcome cross-moment vector as functions of the raw moments v, and write $A = A(\mu)$ and $b = b(\mu)$. These functions are affine in the chosen raw-moment coordinates, so their derivatives of order two and higher vanish. For a perturbation direction h_s of the raw moments, write dA_s and db_s for the corresponding perturbations of A and b. Writing $\Omega = A^{-1}$ and $\theta = \Omega b$, differentiation of $A\theta = b$ gives the exact derivative identities

$$D\theta[h] = \Omega(db - dA\,\theta),$$

$$D^2\theta[h_1, h_2] = -\Omega dA_1 \Omega(db_2 - dA_2\theta) - \Omega dA_2 \Omega(db_1 - dA_1\theta),$$

$$D^3\theta[h_1, h_2, h_3] = \sum_{\varsigma \in \mathfrak{S}_3} \Omega dA_{\varsigma_1} \Omega dA_{\varsigma_2} \Omega(db_{\varsigma_3} - dA_{\varsigma_3}\theta).$$

Here $\mathfrak{S}_3$ is the set of all six permutations $(\varsigma_1, \varsigma_2, \varsigma_3)$ of $(1, 2, 3)$; dA and db in the first identity are the perturbations in direction h.

Step 4: account for the centering in IREG. REG's estimator is the treatment coefficient, $g_{\mathrm{REG}}(v) = \theta_W(v)$. IREG's is $g_{\mathrm{IREG}}(v) = \theta_W(v) + \bar{z}(v)^\top \theta_{WZ}(v)$, where $\bar{z}(v)$ is the full-sample covariate mean, a linear coordinate of v that vanishes at μ in whitened coordinates, and $\theta_{WZ} = \gamma$ at μ. The product rule gives

$$a_{\mathrm{IREG}}[h] = D\theta_W[h] + D\bar{z}[h]^\top \gamma,$$

$$H_{\mathrm{IREG}}[h_1, h_2] = D^2\theta_W[h_1, h_2] + D\bar{z}[h_1]^\top D\theta_{WZ}[h_2] + D\bar{z}[h_2]^\top D\theta_{WZ}[h_1],$$

$$A_{3,\mathrm{IREG}}[h_1, h_2, h_3] = D^3\theta_W[h_1, h_2, h_3] + \sum_{\{i,j,k\} = \{1,2,3\}} D\bar{z}[h_i]^\top D^2\theta_{WZ}[h_j, h_k],$$

where the sum means the three terms with $(i, j, k) = (1, 2, 3), (2, 1, 3), (3, 1, 2)$. The covariance and third-moment blocks of $\mathcal{V}_i$ are products of moments of $(Z_i, \omega_i, \varepsilon_i)$ under the canonical representation in Step 1, with W_i marginally Bernoulli(1/2).

Step 5: evaluate the contractions and take IREG minus REG. Write $\Delta\mathcal{T}_k = \mathcal{T}_{k,\mathrm{IREG}} - \mathcal{T}_{k,\mathrm{REG}}$. Substituting these derivative tensors and the covariance and third-moment blocks of

$\mathcal{V}_i$ into (27)–(29), we reduce the resulting indexed expressions using tensor symmetries, dummy-index relabeling, and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$. This gives

$$\Delta\mathcal{T}_1 = 2G + 2F + 8\Xi_\gamma, \tag{30}$$

$$\Delta\mathcal{T}_2 = -3G - F - Q - \mathcal{R} - 8\Xi_\gamma - 8A_P - 8B_P, \tag{31}$$

$$\Delta\mathcal{T}_3 = -(2p+6)G - 2\mathcal{R} - 8A_P + 8C_P + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top\mathbb{E}[Z\varepsilon^2], \tag{32}$$

where

$$\Xi_\gamma = \mathbb{E}[(\gamma^\top Z)\|Z\|^2\omega\varepsilon]$$

with $\omega = W - \frac{1}{2}$. The accompanying verification script enumerates the full Cartesian products of the expansion directions and reduces the resulting indexed expressions using only tensor symmetries, dummy-index relabeling, and the identities implied by $\mathbb{E}[ZZ^\top] = I_p$ and $\omega_i^2 = 1/4$, with the dimension p kept symbolic. The symbolic remainder in each of the three rows is identically zero; the verification is implemented in `verification/general_p_symbolic_certificate.py` in the replication repository and runs in under a minute.

Step 6: collect the terms. Adding the three rows gives the coefficient of G as $2-3-(2p+6) = -(2p+7)$ and the coefficient of F as $2-1=1$. The Ξ_γ terms cancel. The remaining coefficients are -1 for Q, -3 for $\mathcal{R}$, and $(-16,-8,8)$ for (A_P, B_P, C_P), together with the two residual terms in (26). This proves the stated formula.

One contraction can be checked directly without the full tensor calculation. The third covariance pairing in (28) is

$$\frac{1}{4}(H_{ij}\Sigma_{ij}^{\mathcal{V}})^2.$$

It is the square of the leading bias coefficient. By the bias calculation in Lemma 14, those coefficients at balanced allocation are $b_I = -4\,\mathrm{tr}(P)$ and $b_R = -4\,\mathrm{tr}(P) - \gamma^\top q$. They are also the Bernoulli coefficients: the conditional bias coefficient is smooth in r, and Lemma 7, applied to clipped errors, changes the mean by only $O(n^{-2})$. Conditioning on the treatment count and using its symmetry then preserves the leading bias coefficient. Thus this pairing contributes

$$b_I^2 - b_R^2 = -(\gamma^\top q)^2 - 8(\gamma^\top q)\,\mathrm{tr}(P) = -Q - 8B_P.$$

These are exactly the two bias-squared terms in the second row. The other pairings and the linear–cubic contraction account for the remaining variance contributions.

The computational certificate checks the remaining algebra. Derived by hand and displayed above: the expansion terms (27)–(29), the Gram matrices and coefficients at μ, the derivative identities, the product-rule structure of the IREG centering functional, and the reduction rules. Verified symbolically by the accompanying implementation: the substitution of these objects into (27)–(29) and the reduction of the resulting indexed sums to the rows (30)–(32), with symbolic p and zero remainder. The resulting constant is checked independently by Monte Carlo in Appendix B. □

A.4 Recovering the matrix formula in the main theorem

Adding the $4G$ correction of Theorem 10 to the Bernoulli constant in Theorem 11 changes only the coefficient of G, from $-(2p+7)$ to $-(2p+3)$. Hence

$$D_p = -(2p+3)G + F - Q - 3\mathcal{R} - 16A_P - 8B_P + 8C_P + 8\,\mathrm{tr}(\mathcal{E}^2) + 8S^\top\mathbb{E}[Z\varepsilon^2], \tag{33}$$

which Lemma 12 rewrites in the matrix form of Theorem 2. Together with Lemma 7, this proves its capped-MSE identity.

Lemma 12 (Equivalent matrix form)**.** *With K, S, S_γ, $\mathcal{E}$ and $\mathcal{E}_W$ as in Theorem 2,*

$$G = \|\gamma\|^2, \qquad F = \gamma^\top K\gamma, \qquad Q = (\gamma^\top S)^2, \qquad \mathcal{R} = \|S_\gamma\|_F^2,$$
$$A_P = \operatorname{tr}(S_\gamma\mathcal{E}_W), \qquad B_P = (\gamma^\top S)\operatorname{tr}(\mathcal{E}_W), \qquad C_P = \gamma^\top\mathcal{E}_W S.$$

Proof. First,

$$F = \gamma_i\gamma_j T_{4,ijkl}\delta_{kl} = \gamma^\top\mathbb{E}[ZZ^\top\|Z\|^2]\gamma = \gamma^\top K\gamma,$$

and

$$S = \mathbb{E}[Z_iZ_jZ_j]_{i=1}^p = q$$

gives

$$Q = (\gamma^\top q)^2 = (\gamma^\top S)^2.$$

Next,

$$(S_\gamma)_{ab} = \mathbb{E}[Z_aZ_b(Z^\top\gamma)] = T_{3,abk}\gamma_k,$$

so

$$\|S_\gamma\|_F^2 = \sum_{a,b}(S_\gamma)_{ab}^2 = \gamma_k\gamma_\ell T_{3,abk}T_{3,ab\ell} = \gamma_k\gamma_\ell L_{k\ell} = \mathcal{R}.$$

Since S_γ and $\mathcal{E}_W$ are symmetric,

$$\operatorname{tr}(S_\gamma\mathcal{E}_W) = (S_\gamma)_{jk}(\mathcal{E}_W)_{jk} = \gamma_i T_{3,ijk}P_{jk} = A_P;$$

finally,

$$B_P = (\gamma^\top q)\operatorname{tr}(P) = (\gamma^\top S)\operatorname{tr}(\mathcal{E}_W)$$

and

$$C_P = \gamma^\top Pq = \gamma^\top\mathcal{E}_W S.$$

□

A.5 An exact estimator difference and the leading biases

The next identity explains both the first-order equivalence of REG and IREG and the control variate used in the numerical verification. Their difference is a product of covariate imbalance, scatter-matrix imbalance, and the difference between the fitted arm slopes.

We begin with an exact identity used by the variance-reduced verification of Appendix B.

Lemma 13 (Exact difference identity)**.** *Let $\bar{X}_w$ and*

$$\mathcal{S}_w = \sum_{i:W_i=w}(X_i - \bar{X}_w)(X_i - \bar{X}_w)^\top$$

denote the within-arm means and scatter matrices, let

$$\hat{\beta}_w = \mathcal{S}_w^{-1}\sum_{i:W_i=w}(X_i - \bar{X}_w)(Y_i - \bar{Y}_w),$$

and write $d_X = \bar{X}_1 - \bar{X}_0$. Then, whenever $\mathcal{S}_0$, $\mathcal{S}_1$ and $\mathcal{S}_0 + \mathcal{S}_1$ are invertible and $n_1 = n_0$,

$$\hat{\tau}_{\mathrm{REG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top(\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1\hat{\beta}_1 + \mathcal{S}_0\hat{\beta}_0),$$

$$\hat{\tau}_{\mathrm{IREG}} = \bar{Y}_1 - \bar{Y}_0 - d_X^\top\frac{\hat{\beta}_1 + \hat{\beta}_0}{2},$$

and consequently

$$\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}} = \tfrac{1}{2}\, d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1 - \mathcal{S}_0)(\hat{\beta}_1 - \hat{\beta}_0) = O_P(n^{-1}).$$

Here the algebraic identity is exact on the stated invertibility event; the stochastic order uses the sampling and moment conditions of Theorem 2. The probability of invertibility tends to one, since the normalized within-arm scatter matrices converge to $\Sigma \succ 0$.

Proof. Both expressions for the estimators are the Frisch–Waugh–Lovell representations of (2): the coefficient on X in the REG regression is the pooled within-arm slope $(\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1\hat{\beta}_1 + \mathcal{S}_0\hat{\beta}_0)$, and IREG fits the two arms separately, so with $n_1 = n_0$ the overall covariate mean satisfies

$$\bar{X}_1 - \bar{X} = d_X/2 = -(\bar{X}_0 - \bar{X}).$$

Subtracting,

$$\begin{aligned}\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}} &= d_X^\top \left[(\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1\hat{\beta}_1 + \mathcal{S}_0\hat{\beta}_0) - \tfrac{1}{2}(\hat{\beta}_1 + \hat{\beta}_0)\right] \\ &= \tfrac{1}{2} d_X^\top (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1 - \mathcal{S}_0)(\hat{\beta}_1 - \hat{\beta}_0).\end{aligned}$$

The stochastic order follows from

$$d_X = O_P(n^{-1/2}), \qquad (\mathcal{S}_1 + \mathcal{S}_0)^{-1}(\mathcal{S}_1 - \mathcal{S}_0) = O_P(n^{-1/2}), \qquad \hat{\beta}_1 - \hat{\beta}_0 = O_P(1).$$

□

In whitened coordinates, the same Taylor expansion gives

$$\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}} = \tfrac{1}{4} d_Z^\top (A_1 - A_0)\gamma + O_P(n^{-3/2}),$$

where $d_Z = \bar{Z}_1 - \bar{Z}_0$ and A_w is the arm-w mean of $ZZ^\top - I_p$; the arm sizes here remain $n_1 = n_0 = n/2$. Indeed, replacing each centered scatter matrix by its uncentered second-moment version changes its normalized value by $O_P(n^{-1})$; the inverse-Gram and slope expansions each contribute another $O_P(n^{-1/2})$ correction to their leading terms. Multiplication by $d_Z = O_P(n^{-1/2})$ gives the stated remainder. The leading quadratic term is used in the control variate of (34) below.

The final lemma separates the leading biases of the two estimators. Both share the arm-specific residual contribution; REG has the additional contribution $-\gamma^\top S/n$ from the interaction of covariate skewness and treatment-effect heterogeneity.

Lemma 14 (Leading clipped-error bias)**.** *Under the sampling and moment conditions of Theorem 2 and balanced complete randomization, for* $e_{j,c}$ *as in Remark 3,*

$$\mathbb{E}[e_{\mathrm{IREG},c}] = -\frac{4\,\mathrm{tr}(\mathcal{E}_W)}{n} + O(n^{-2}),$$

$$\mathbb{E}[e_{\mathrm{REG},c}] = -\frac{4\,\mathrm{tr}(\mathcal{E}_W) + \gamma^\top S}{n} + O(n^{-2}).$$

Proof. By Lemma 7, it suffices to calculate the cubic Taylor-polynomial biases up to $O(n^{-2})$; no expectation of an inverse sample Gram matrix is required. First suppose n is even and set $m = n/2$. In whitened coordinates, define

$$d_Z = \bar{Z}_1 - \bar{Z}_0, \qquad t_w = \frac{1}{m}\sum_{i:W_i=w} Z_i\varepsilon_i(w), \qquad A_w = \frac{1}{m}\sum_{i:W_i=w}(Z_iZ_i^\top - I_p).$$

Each is a centered sample-mean quantity. Write $b_w = \Sigma^{1/2}\beta^*_{(w)}$ for the population arm slope in whitened coordinates, so $b_1 - b_0 = \gamma$. The common linear error term is

$$L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0, \qquad \mathbb{E}[L] = 0.$$

Here $\bar{Z} = (\bar{Z}_1 + \bar{Z}_0)/2$ and $\bar{\varepsilon}_w = m^{-1}\sum_{i:W_i=w} \varepsilon_i(w)$.

Step 1: identify the quadratic terms. The linear Taylor term of an arm slope is t_w. The linear Taylor term of the pooled slope, expanded around $(b_1 + b_0)/2$, is

$$\frac{t_1 + t_0}{2} + \frac{1}{4}(A_1 - A_0)\gamma.$$

To verify the latter expression, expand the pooled normal equations: the first perturbation of the within-arm Gram matrix is $(A_1 + A_0)/2$, whereas its cross-moment perturbation includes $(A_1 b_1 + A_0 b_0)/2 + (t_1 + t_0)/2$. Subtracting the Gram perturbation times $(b_1 + b_0)/2$ leaves the displayed expression. Products of arm means enter only at degree two in the slope and hence at degree three in the treatment estimator.

The quadratic estimator terms are therefore

$$q_{2,I} = -\frac{1}{2} d_Z^\top (t_1 + t_0), \qquad q_{2,R} = -d_Z^\top \left\{ \frac{t_1 + t_0}{2} + \frac{1}{4}(A_1 - A_0)\gamma \right\}.$$

The cubic Taylor terms have expectations $O(n^{-2})$, because a product of three centered sample means has expectation of that order. Thus the leading expansion biases equal $\mathbb{E}[q_{2,I}]$ and $\mathbb{E}[q_{2,R}]$, respectively, with remainder $O(n^{-2})$.

Step 2: evaluate the residual contribution. Independence of the arms and centering give the exact identities

$$\mathbb{E}[d_Z^\top t_1] = \frac{1}{m}\mathbb{E}[\|Z\|^2 \varepsilon(1)], \qquad \mathbb{E}[d_Z^\top t_0] = -\frac{1}{m}\mathbb{E}[\|Z\|^2 \varepsilon(0)].$$

Consequently,

$$\mathbb{E}[q_{2,I}] = -\frac{1}{2m}\operatorname{tr}\left\{\mathbb{E}[ZZ^\top \varepsilon(1)] - \mathbb{E}[ZZ^\top \varepsilon(0)]\right\} = -\frac{4\operatorname{tr}(\mathcal{E}_W)}{n}.$$

Step 3: evaluate REG's additional heterogeneity contribution. For either arm,

$$\mathbb{E}[\bar{Z}_w^\top A_w \gamma] = \frac{1}{m}\mathbb{E}[Z^\top (ZZ^\top - I_p)\gamma] = \frac{\gamma^\top S}{m}.$$

The cross-arm terms vanish, so

$$\mathbb{E}[d_Z^\top (A_1 - A_0)\gamma] = \frac{2\gamma^\top S}{m}.$$

Multiplication by $-1/4$ adds $-\gamma^\top S/n$ to REG's bias, proving both formulas for even n.

For odd n, the arm fraction differs from $1/2$ by $O(n^{-1})$. The leading quadratic expectation is n^{-1} times a smooth function of that fraction, by the same moment calculation as in Lemma 8. Its change is therefore $O(n^{-2})$. The formulas hold for odd n with the same stated remainder. □

B Numerical illustration of Theorem 2

The symbolic certificate verifies the algebra of Theorem 11 exactly. Simulation serves a different purpose: it assesses whether the fitted estimators approach the predicted constant in the chosen designs and whether the experiment can distinguish that constant from plausible alternatives, including an omitted CRT correction. These checks provide numerical evidence; they do not replace the proof. The simulations below use ordinary, uncapped squared errors, whereas Theorem 2 concerns capped MSE. Additional tail control is needed to transfer the expansion to ordinary MSE, as discussed in Appendix A.

Exact targets. We use data-generating processes with whitened covariates having independent coordinates with known moments. The oracle residuals take the form

$$\varepsilon(w) = g_w(Z) + s_w(Z)\eta,$$

where η is independent of Z with mean zero and unit variance, g_w is a polynomial orthogonal to $(1, Z)$, and s_w^2 is a polynomial. Each term in D_p is therefore a finite combination of population moments and can be computed in exact rational arithmetic. The code also checks $\mathbb{E}[g_w] = \mathbb{E}[Zg_w] = 0$, so that violations of the orthogonality conditions cannot pass unnoticed.

We use six designs, chosen so that every nontrivial block of D_p is active in at least one design: Gaussian homoskedastic covariates; centered-exponential covariates with heterogeneity; centered-exponential covariates with heteroskedasticity and $\gamma = 0$; centered-exponential covariates with heterogeneity, arm-specific misspecification, and arm-specific heteroskedasticity; Gaussian covariates with arm-specific misspecification and heteroskedasticity; and a mixed skewed/Gaussian design. These designs are analytically tractable and differ from the inference designs of Section 3; their exact definitions are in `simulations/dgps.py` of the replication repository.

Variance reduction for numerical verification. Our simulation estimates the finite-sample quantity

$$D_n = n^2\{\mathrm{MSE}(\hat{\tau}_{\mathrm{IREG}}) - \mathrm{MSE}(\hat{\tau}_{\mathrm{REG}})\}.$$

Under the additional mean-square remainder conditions discussed in Appendix A, D_n converges to the expansion-MSE constant D_p. Comparing the simulated D_n with D_p therefore checks how closely the fitted estimators follow the second-order prediction in the chosen designs.

Suppose we generate B independent datasets, each with even sample size n and $n/2$ units per arm. On replication b, let $e_{j,b} = \hat{\tau}_{j,b} - \tau$ and define

$$T_b = n^2(e_{\mathrm{IREG},b}^2 - e_{\mathrm{REG},b}^2).$$

When the fitted estimators are defined and have finite second moments, the average of the T_b is unbiased for D_n. However, individual values of T_b can fluctuate substantially, making this direct Monte Carlo estimate noisy.

To identify the leading source of this noise, write $d_Z = \bar{Z}_1 - \bar{Z}_0$, let A_w be the arm-w mean of $ZZ^\top - I_p$, and let $\bar{\varepsilon}_w$ be the arm-w mean of the oracle residuals. Define

$$d_1 = \tfrac{1}{4} d_Z^\top (A_1 - A_0)\gamma, \qquad L = \gamma^\top \bar{Z} + \bar{\varepsilon}_1 - \bar{\varepsilon}_0.$$

By Lemma 13, d_1 is the leading term of $\hat{\tau}_{\mathrm{IREG}} - \hat{\tau}_{\mathrm{REG}}$, while L is the common first-order estimation error. Thus

$$\begin{aligned} e_{\mathrm{IREG}}^2 - e_{\mathrm{REG}}^2 &= (e_{\mathrm{IREG}} - e_{\mathrm{REG}})(e_{\mathrm{IREG}} + e_{\mathrm{REG}}) \\ &= 2d_1 L + O_P(n^{-2}). \end{aligned}$$

Since $d_1 = O_P(n^{-1})$ and $L = O_P(n^{-1/2})$, the term $2n^2 d_1 L$ is $O_P(\sqrt{n})$. Under the sampling and moment conditions of Theorem 2, its variance is $O(n)$. This term can therefore produce large fluctuations in T_b, even though the quantity we want to estimate remains bounded as n grows.

We reduce this noise using a control variate. On each replication, set

$$C_b = 2n^2 d_{1,b} L_b.$$

The quantity C_b captures the leading fluctuation in T_b. Subtracting its centered version gives the adjusted estimate

$$\hat{D}_p = \frac{1}{B} \sum_{b=1}^{B} \{T_b - (C_b - \mathbb{E}[C_b])\}.$$

Because $\mathbb{E}[C_b - \mathbb{E}[C_b]] = 0$, this adjustment leaves the expectation unchanged: $\mathbb{E}[\hat{D}_p] = D_n$. At a fixed n, $\hat{D}_p$ estimates the scaled finite-sample MSE difference, rather than being unbiased for the limiting constant D_p.

The control mean is available analytically because the simulation distribution is known. For an arm of size $m = n/2$, the expectation of a product of three centered IID sample means equals the corresponding one-unit third joint cumulant divided by m^2. Applying this identity gives

$$\mathbb{E}[d_1 L] = \frac{1}{n^2} \mathbb{E}\Big[(\|Z\|^2 - 1)(\gamma^\top Z)\{\gamma^\top Z + \varepsilon(1) - \varepsilon(0)\}\Big], \tag{34}$$

and hence $\mathbb{E}[C_b] = 2n^2 \mathbb{E}[d_1 L]$. Only this control mean is computed analytically; the remaining contribution is estimated from the simulated datasets.

The adjustment removes the leading fluctuation in probability:

$$T_b - C_b = n^2 \{e^2_{\mathrm{IREG},b} - e^2_{\mathrm{REG},b} - 2d_{1,b} L_b\} = O_P(1).$$

This stochastic-order statement alone does not guarantee bounded variance. A sufficient additional condition is

$$\sup_n \mathbb{E}[(T_b - C_b)^2] < \infty.$$

Under this condition, the adjusted per-replication variance is $O(1)$, and the Monte Carlo standard error of $\hat{D}_p$ is $O(B^{-1/2})$ uniformly in n. The sampling and moment conditions of Theorem 2 alone do not establish this condition, because the fitted estimators involve inverse sample Gram matrices. The standard errors reported below are empirical simulation summaries.

The replication code reports both the raw and adjusted estimates. In the Gaussian design, the raw per-replication standard deviation increases from 151 at $n = 200$ to 293 at $n = 800$, while the control-variate standard deviation is 56 and 55, respectively. At $n = 3200$, the control-variate Monte Carlo standard error is ten times smaller than that of the raw statistic.

Results. Table 3 reports the exact targets and the control-variate estimates at $n \in \{200, 800, 3200\}$, using $B = 400{,}000$ replications for the two smaller sample sizes and $B = 120{,}000$ for $n = 3200$. In these simulations, the control variate substantially reduces the observed per-replication standard deviation, making the estimated Monte Carlo errors small enough to assess finite-n discrepancies.

The largest discrepancy occurs in the all-blocks design (fourth row). There the estimate is 32.40 at $n = 200$, compared with the target 44.52, a difference of 12.1 (about 22 standard errors). The difference falls to 5.7 at $n = 800$ and 2.55 at $n = 3200$. The Bernoulli constant $D_{\mathrm{Bern}} = D_p - 4G = 41.32$ is within one standard error of the $n = 3200$ estimate, so that sample

size alone does not distinguish D_{Bern} from D_p in this design. The successive increments, 6.4 and 3.1, shrink by a factor of about two as n quadruples. A leading remainder proportional to n^{-3} in MSE would shrink these increments by a factor of four. The observed increments do not establish a slower asymptotic rate: these sample sizes may be outside that regime, and the simulations use uncapped MSE. We do not extrapolate. The ablation in Table 4 separates D_p from D_{Bern} more clearly in other designs.

At $n = 3200$, all six estimates are within 2.6 standard errors of their targets. The two largest deviations are $z = -2.6$ in the all-blocks design and $z = +2.3$ in the $\gamma = 0$ heteroskedastic design. In the latter, the estimate moves from 29.82 at $n = 200$ and 31.73 at $n = 800$ to 33.25 at $n = 3200$. With only six designs, deviations of this size are compatible with Monte Carlo error. The results are consistent with a remainder that is smaller than the n^{-2} leading term, but three sample sizes are not enough to estimate its rate.

Table 3: Simulation comparison with Theorem 2. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses. $B = 400{,}000$ for $n \in \{200, 800\}$ and $B = 120{,}000$ for $n = 3200$. Parentheses after a design name list the blocks of D_p that are active, in the notation of Theorem 11 and Lemma 12 ($F = \gamma^\top K\gamma$, $Q = (\gamma^\top S)^2$, $\mathcal{R} = \|S_\gamma\|_F^2$, $E = \mathcal{E}$, $E_W = \mathcal{E}_W$); the mixed design has heterogeneity along its skewed coordinate and heteroskedastic residuals, so the covariate-shape blocks and $8S^\top\mathbb{E}[Z\varepsilon^2]$ are active.

DGP (active blocks)	D_p (exact)	$n = 200$	$n = 800$	$n = 3200$
Gaussian, homoskedastic, $p = 3$ (F)	-5.00	-4.70 (0.09)	-4.99 (0.09)	-5.17 (0.16)
Exponential covariates, homoskedastic (F, Q, R)	-20.25	-17.17 (0.21)	-19.60 (0.22)	-19.75 (0.39)
Exponential, heteroskedastic, $\gamma = 0$	32.00	29.82 (0.28)	31.73 (0.29)	33.25 (0.54)
Exponential, HTE + arm-specific misspec. (all)	44.52	32.40 (0.56)	38.82 (0.55)	41.97 (0.98)
Gaussian, arm-specific misspec. ($\operatorname{tr} E^2, E_W$)	-2.32	-2.36 (0.09)	-2.34 (0.09)	-2.45 (0.16)
Mixed skewed/Gaussian, heteroskedastic	28.00	25.94 (0.27)	27.85 (0.28)	28.27 (0.53)

Power: what the verification would reject. Agreement with D_p is more informative when the simulation can also detect errors in the formula. We therefore compare the estimates with several deliberately perturbed candidate constants. Table 4 reports z-statistics at $n = 3200$ for the exact D_p and several perturbed formulas: the Bernoulli constant $D_{\text{Bern}} = D_p - 4G$, which omits the CRT correction; the Gaussian special case $-(p+1)G$; and versions of D_p with individual blocks removed.

The exact formula has $|z| \le 2.6$ in every design, whereas each alternative has $|z| > 3$ in at least one design, in some cases by tens of standard errors. The check has limited power when an alternative differs from D_p by less than the residual finite-n bias. In the all-blocks design, for example, $D_{\text{Bern}} = D_p - 3.2$ is not separated from D_p at $n = 3200$, which is why multiple designs are useful.

Without the control variate, the standard errors at $n = 3200$ would be 1.5 to 3.8, four to ten times larger in the heterogeneous designs. The Bernoulli constant would then not be separated from D_p in any design ($|z| \le 3$ throughout), so the simulation would have little power to detect an omitted CRT correction.

Further covariate families and dimensions. The six designs above have $p \le 3$ and independent coordinates. We also consider four covariate families at $p \in \{2, 5, 8\}$, including dependent coordinates (Table 5).

For Gaussian covariates, $D_p = -(p+1)G$. For Rademacher covariates, $K = pI_p$ and $D_p = -(p+3)G$. For a symmetric scale mixture of normals, where 2% of units have six times

Table 4: Ablation: $z = (\hat{D}_p - D^{\text{alt}})/\widehat{\text{SE}}$ at $n = 3200$ for each design and candidate constant. Values with $|z| > 3$ indicate rejection of that candidate. Each "drop" column removes one block from D_p: the term $-3\mathcal{R}$, the term $-Q$, the residual terms $8\,\text{tr}(\mathcal{E}^2) + 8S^\top \mathbb{E}[Z\varepsilon^2]$ ("resid."), or the three terms involving $\mathcal{E}_W$.

DGP	D_p	$D_p - 4G$	$-(p+1)G$	drop $-3\mathcal{R}$	drop $-Q$	drop resid.	drop $\mathcal{E}_W$
Gaussian, homoskedastic, $p = 3$ (F)	-1.0	$+29.8$	-1.0	-1.0	-1.0	-1.0	-1.0
Exponential covariates, homoskedastic (F, Q, R)	$+1.3$	$+14.2$	-41.3	-37.4	-21.9	$+1.3$	$+1.3$
Exponential, heteroskedastic, $\gamma = 0$	$+2.3$	$+2.3$	$+61.1$	$+2.3$	$+2.3$	$+61.1$	$+2.3$
Exponential, HTE + arm-specific misspec. (all)	-2.6	$+0.7$	$+45.1$	-12.4	-3.2	$+65.8$	-17.7
Gaussian, arm-specific misspec. ($\text{tr}\,E^2, E_W$)	-0.8	$+23.5$	$+3.4$	-0.8	-0.8	$+3.4$	-0.8
Mixed skewed/Gaussian, heteroskedastic	$+0.5$	$+4.2$	$+55.7$	-5.1	-1.4	$+60.3$	$+0.5$

the standard deviation, write s for the mixing scale ($s = 6$ with probability 0.02 and $s = 1$ otherwise). After standardizing the mixture,

$$K = (p+2)\tilde{\kappa} I_p, \qquad \tilde{\kappa} = \frac{\mathbb{E}[s^4]}{\mathbb{E}[s^2]^2},$$

and

$$D_p = G\{(p+2)\tilde{\kappa} - (2p+3)\}.$$

For Gaussian covariates with quadratic misspecification $c(X^\top X - \text{tr}\,\Sigma)$, where $c = 0.3$ and $\Sigma_{jk} = 0.3^{|j-k|}$,

$$\mathcal{E} = 2c\Sigma, \qquad D_p = -(p+1)G + 32c^2\,\text{tr}(\Sigma^2).$$

Every family uses homoskedastic $N(0,1)$ noise shared by the potential outcomes, with $G = 0.49$ (1.96 for the Rademacher family). Because both estimators are invariant to invertible linear transformations of the covariates, we simulate whitened covariates; dependence enters through γ and, for the misspecified family, through $\text{tr}(\Sigma^2)$.

The targets are exact. Recomputing them from the matrix form of Theorem 2 using 10^6 Monte Carlo draws reproduces them to within 5%, with the largest discrepancy in the scale mixture, where fourth moments are harder to estimate. Using exact targets avoids this additional Monte Carlo error.

With $n = 4{,}000$ and $B = 20{,}000$ replications per design, every estimate is within 1.55 standard errors of its target, and the control variate reduces the standard errors by factors of 3.9 to 11.4 relative to the raw estimator on the same replications.

Figure 1 shows the z-statistics for all 18 checks: the designs of Table 3 at $n = 3200$ and those of Table 5. Of these, 16 are within ± 1.96; the two exceptions are among the deviations discussed above.

Figure 2 shows the estimates as a function of n. For the four families at $p = 5$, the standard errors remain roughly constant as n grows. One of the twelve points below $n = 4{,}000$ lies outside ± 1.96 standard errors (the scale mixture at $n = 1{,}000$, $z = -2.21$), while the neighboring points at $n = 500$ and $n = 2{,}000$ are within 0.6 standard errors. With twenty-four family checks, an isolated deviation of this size is compatible with Monte Carlo variation.

The CRT correction and the bias decomposition. We check Theorem 10 separately by estimating the n^{-2} constant under both balanced CRT and Bernoulli(1/2) assignment on the same designs (`verification/crt_correction_certificate.py`). In the Gaussian design, the estimated gap between the two designs is 4.3 ± 0.6 at $n = 200$ and 6.3 ± 1.6 at $n = 800$, compared with the predicted $4G = 5$. In the exponential-covariate design, the corresponding gaps are 3.0 ± 1.1 and 8.6 ± 3.0. All four comparisons have $|z| < 2$.

Table 5: Further simulation comparison with Theorem 2: four covariate families at $p \in \{2, 5, 8\}$, $n = 4{,}000$, $B = 20{,}000$. Targets are exact; estimates are the control-variate estimator with Monte Carlo standard errors in parentheses; the last column is the standard error of the raw estimator on the same replications.

Family	p	D_p (exact)	Estimate	z	Raw SE
Gaussian	2	-1.47	-1.31 (0.20)	$+0.82$	2.27
Gaussian	5	-2.94	-2.57 (0.42)	$+0.90$	3.22
Gaussian	8	-4.41	-3.50 (0.62)	$+1.48$	4.00
Scale mixture	2	14.81	13.33 (1.06)	-1.40	8.14
Scale mixture	5	25.56	26.51 (1.83)	$+0.52$	10.39
Scale mixture	8	36.30	35.37 (2.74)	-0.34	12.85
Rademacher	2	-9.80	-9.51 (0.29)	$+1.02$	2.97
Rademacher	5	-15.68	-16.65 (0.62)	-1.55	6.20
Rademacher	8	-21.56	-20.94 (0.94)	$+0.67$	8.13
Quadratic misspec.	2	4.81	4.75 (0.35)	-0.17	3.12
Quadratic misspec.	5	13.68	14.10 (0.96)	$+0.44$	4.99
Quadratic misspec.	8	22.56	22.19 (1.72)	-0.21	6.77

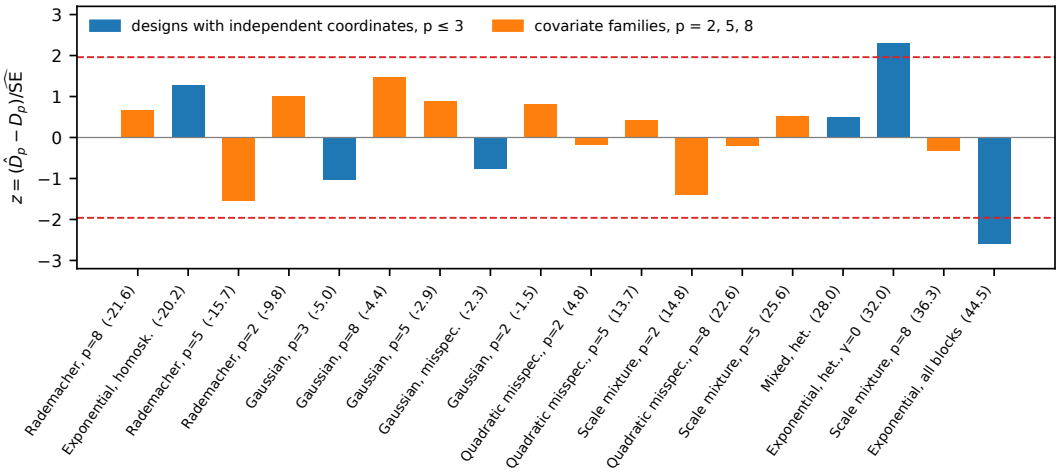


Figure 1: z-statistics $(\hat{D}_p - D_p)/\widehat{\mathrm{SE}}$ of the control-variate estimates for the designs of Table 3 ($n = 3{,}200$) and Table 5 ($n = 4{,}000$), sorted by D_p (in parentheses). Dashed lines mark ± 1.96.

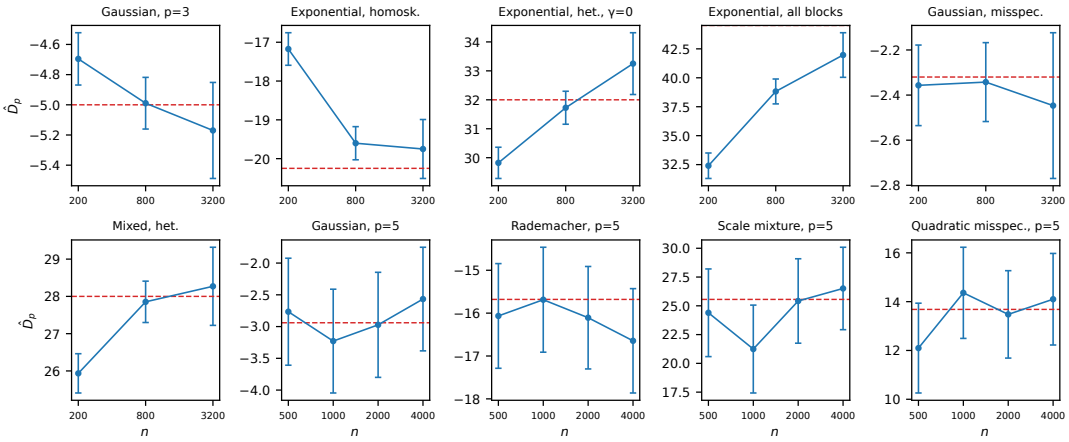


Figure 2: Control-variate estimates of D_p against n (log scale) with ± 1.96 standard-error bars; the dashed line is the exact D_p. The first six panels are the designs of Table 3 ($n = 200$, 800, 3,200); the last four are the families of Table 5 at $p = 5$ ($n = 500$ to 4,000).

This check uses the raw statistic because the control variate is derived for the two-independent-samples representation of the CRT, so its power is limited. The ablation in Table 4 provides a sharper test: there, the Bernoulli constant is rejected by as many as thirty standard errors.

We also compare the leading coefficients in Remark 3 with simulations of the ordinary, unclipped biases of both estimators (`simulations/run_bias.py`). With 10^6 replications, at $n = 800$ all six estimated biases (REG and IREG in three designs) are within 0.8 Monte Carlo standard errors of their leading-order predictions. At $n = 200$, the all-blocks design differs by about 11% of the predicted REG bias ($z = 9.3$) and by 5% of the predicted IREG bias ($z = 3.5$); by $n = 800$, both differences are negligible at the reported Monte Carlo precision ($z = 0.4$ and $z = -0.5$), consistent with the leading bias coefficients. Lemma 14 proves the $O(n^{-2})$ remainder for clipped errors; the ordinary-bias simulation does not establish that bound for unclipped errors.

C Range of validity of the second-order approximation

Theorem 2 is an asymptotic result. A separate question is when the approximation

$$1 + \Delta/n$$

is accurate for the finite-sample MSE ratio. We examine this in the scalar lognormal family of the introduction, with $\sigma(X) = 1 + X$, now with a treatment-effect slope of 0.4 fixed while $X \sim \text{LogNormal}(0, s^2)$ varies. The ratio compares IREG with REG, so values above one favor REG. As s increases, the covariate tails become heavier and the higher moments entering Δ grow rapidly. We ask whether those population moments describe the MSE comparison accurately at the sample sizes used in the simulations.

For $s = 0.8$ (skewness 3.7), the simulated excess MSE is 75% of the predicted excess Δ/n at $n = 500$ and 85% at $n = 2{,}000$; for $s = 0.6$, the corresponding figures are 91% and 94%. The approximation deteriorates as the tails become heavier. At $s = 1.5$, for example, the predicted

Table 6: Simulated MSE ratio $\widehat{\mathrm{MSE}}_{\mathrm{IREG}}/\widehat{\mathrm{MSE}}_{\mathrm{REG}}$ versus the second-order prediction $1+\Delta/n$, for $X \sim \mathrm{LogNormal}(0, s^2)$, $\sigma(X) = 1 + X$, treatment-effect slope 0.4. Monte Carlo standard errors of the simulated ratios are given in parentheses in units of the last reported digit. The introductory example is this family at $s = 1.1$ with no treatment-effect heterogeneity; the heavy-tailed example ($s = 1.8$, $\sigma(X) = 1 + 1.5X$, slope 0.3) is a separate design.

			$n = 500$		$n = 2000$		$n = 8000$	
s	skew Z	Δ	sim.	pred.	sim.	pred.	sim.	pred.
0.6	2.3	4	1.0072 (1)	1.0079	1.0019 (1)	1.0020	1.0004 (1)	1.0005
0.8	3.7	12	1.0184 (3)	1.0243	1.0052 (2)	1.0061	1.0014 (1)	1.0015
1.0	6.2	41	1.0429 (7)	1.0827	1.0137 (4)	1.0207	1.0043 (2)	1.0052
1.2	11.2	166	1.0923 (15)	1.3323	1.0347 (11)	1.0831	1.0127 (8)	1.0208
1.5	33.5	2025	1.2517 (51)	5.0503	1.1163 (44)	2.0126	1.0534 (36)	1.2531

ratio is four times the simulated ratio at $n = 500$. We do not interpret this as a universal threshold: the point is that a large deficiency can coexist with substantial finite-sample error in the second-order approximation.

The heavy-tailed example illustrates the same phenomenon more sharply. With $s = 1.8$, $\sigma(X) = 1 + 1.5X$, and treatment-effect slope 0.3, we have $\Delta \approx 3.9 \times 10^4$. The second-order prediction is then 78.5, 20.4, and 5.8 at $n = 500$, 2,000, and 8,000, compared with simulated MSE ratios of 1.68, 1.34, and 1.19. The sign of the comparison is correct, but $1 + \Delta/n$ is no longer a useful numerical approximation to the MSE ratio. In the same design, $R^2_\tau \approx 0.009$, and the simulated coverage of IREG intervals without the plug-in correction is 85.6% with HC1, 88.9% with HC2, and 94.0% with HC3, compared with 94.4% for REG+HC1 and 96.7% for REG+HC3; with the plug-in, IREG+HC1 covers 94.3% (Table 1).

Tail sensitivity. The deficiency is a population-moment calculation. If extreme values account for much of those moments, the asymptotic constant can be large even when samples of the sizes considered seldom reach that part of the distribution. The largest discrepancies occur in precisely these tail-dominated designs. Table 7 recomputes $1 + \Delta/n$ at $n = 500$ after truncating the lognormal law at several high quantiles; the truncated moments are available in closed form. Truncating at the $1-10^{-4}$ quantile reduces Δ by one to three orders of magnitude, showing how strongly the deficiency depends on extreme covariate values. Even truncation at the $1 - 10^{-8}$ quantile still gives predictions several times larger than the simulated ratios.

In these designs, the problem begins already at first order. For the same family without treatment-effect heterogeneity ($\gamma = 0$, the introductory design at other values of s), at $s = 0.6$ the simulated standard deviation of $\hat{\tau}_{\mathrm{REG}}$ is close to the first-order benchmark $\sqrt{V^*/n}$, 0.21 versus 0.21. At $s = 1.5$ and $s = 2$, the corresponding values are 0.72 versus 0.88 and 2.46 versus 4.90, respectively. Thus the heavy-tailed designs are outside the regime where the perturbative expansion gives an accurate quantitative approximation. Monte Carlo error is a secondary issue in the heaviest rows, where rare draws contribute substantially to the simulated MSE. The sign of the REG–IREG comparison agrees with the second-order calculation in every design.

D Designs of the inference simulations

This section specifies the data-generating processes used for the inference comparisons. These are separate from the analytically tractable verification designs of Appendix B. Each compar-

Table 7: Second-order prediction $1 + \Delta/n$ at $n = 500$ for the full lognormal law and for its truncation at the q quantile, against the simulated MSE ratio (Monte Carlo standard error in parentheses, in units of the last digit). The last row is the family of Table 6 at $s = 2$ without treatment-effect heterogeneity ($\gamma = 0$).

Design	Simulated ratio	$1 + \Delta/n$ full law	$q = 1 - 10^{-4}$	$q = 1 - 10^{-6}$	$q = 1 - 10^{-8}$
Heavy-tailed example ($s = 1.8$)	1.68 (3)	78.5	1.90	7.94	26.6
Table 6, $s = 1.5$	1.25 (1)	5.05	1.36	2.44	3.81
$\gamma = 0$, $s = 2$	2.00 (6)	672.1	2.45	17.3	97.4

ison uses shared simulated datasets so that differences across interval procedures reflect the procedure rather than a different simulation draw.

The designs of Tables 1 and 2 all have $n = 500$, balanced complete randomization and 20,000 replications, and every interval is computed on the same simulated datasets. The scalar lognormal designs extend the family of the introduction:

$$p = 1, \qquad X_i \sim \text{LogNormal}(0, s^2), \qquad u_i \sim N(0,1),$$
$$Y_i(0) = 0.9X_i + \sigma(X_i)u_i, \qquad Y_i(1) = Y_i(0) + 1 + gX_i,$$

with $\sigma(X) = 1 + X$ except in the $s = 1.8$ design, where $\sigma(X) = 1 + 1.5X$, and

$$(s, g) \in \{(1.2, 0.3), (1.5, 0.1), (1.5, 0.5), (1.8, 0.3), (2, 0.1), (1.5, 1), (1.5, 3)\}.$$

The remaining designs have $p = 5$, potential outcomes that share their noise, and

$$Y_i(0) = m(X_i) + \sigma(X_i)u_i, \qquad Y_i(1) = Y_i(0) + 1 + \gamma^* X_{i1}, \qquad u_i \sim N(0,1),$$

with $m(x) = 0.9 \sum_j x_j$ unless stated. They differ in the covariate law, the noise scale and the slope γ^*:

- *Gaussian*: $X_i \sim N(0, I_5)$, $\sigma \equiv 1$ and $\gamma^* = 0.3$; the large-heterogeneity Gaussian row instead has $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$ and $\gamma^* = 3$.
- *Misspecified, $G = 0$*: $X_i \sim N(0, \Sigma)$ with $\Sigma_{jk} = 0.5^{|j-k|}$, $\sigma \equiv 1$, $\gamma^* = 0$, and $m(x) = 0.9 \sum_j x_j + 0.20(x_1^2 - 1) + 0.15 x_1 x_2$, so that the working model is misspecified in both arms.
- *Skewed, het., $G = 0$*: independent coordinates, the first a centered LogNormal$(0, 1.1^2)$ and the others standard normal, mixed by the Cholesky factor of the matrix $0.35^{|j-k|}$ and then standardized to unit variances; $\sigma(X) = 1 + 0.5|X_{i1}| + 0.25\|X_i\|$ and $\gamma^* = 0$.
- *Heavy-tailed, het.*: the same construction with LogNormal$(0, 1.5^2)$ and $\sigma(X) = 1 + 0.75|X_{i1}| + 0.25\|X_i\|$; $\gamma^* = 0.5$ in the small-heterogeneity row and $\gamma^* = 3$ in the large-heterogeneity row.
- *Rare extreme values*: $X_{i1} = \sqrt{(1-\rho)/\rho}$ with probability $\rho = 0.02$ and $-\sqrt{\rho/(1-\rho)}$ otherwise, $X_{i2}, \ldots, X_{i5} \sim N(0,1)$, $\sigma \equiv 1$ and $\gamma^* = 1$. With 250 units per arm, an arm contains no rare unit with probability about 0.6%, in which case the interacted design is rank deficient and IREG gives no interval (1.3% of replications), and exactly one rare unit, which then has leverage one in its arm's fit, with probability about 3%.